

MODELING AND ANALYSIS OF A MOVING CONDUCTIVE STRING IN A MAGNETIC FIELD

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(ABSTRACT)

A wide range of physical systems are modeled as axially moving strings; such examples are belts, tapes, wires and fibers with applied electromagnetic fields. In this study, we propose a model that describes the motion of a current-carrying conductive string in a lateral magnetic field, while it is being pulled axially. This model is a generalization of past studies that have neglected one or more properties featured in our system. It is assumed that the string is moving with a constant velocity between two rings that are a finite distance apart. Directions of the magnetic field and the motion of the string coincide. The problem is first considered in a static setting. Stability critical values of the magnetic field, pulling speed, and current are shown to exist when the uniform motion (along a string line) of the string buckles into spiral forms. In the dynamic setting, conditions for stability of certain solutions are presented and discussed. It is shown that there is a divergence between the critical values in the linear dynamic and static cases. Furthermore, traveling wave solutions are examined for certain cases of our general system. We develop an approximate solution for a nonlinear moving string when a periodic nonstationary current flows through the string. Domains of parameters are defined when the string falls into a pre-chaotic state, i.e., the frequency of vibrations is doubled.

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(GENERAL AUDIENCE ABSTRACT)

The modeling and analysis of elastic conductors has applications in areas ranging from manufacturing to particle physics. In this study, we model the motion of a conductive string being pulled (between two rings) while a magnetic field is applied in the lateral direction. This system's stability is categorized through certain parameters such as the applied magnetic field, speed of pulling, and current flowing through the string. The equilibrium states are also analyzed. When the string has a periodic current, approximate solutions (string shape/orientation) are computed. In this case, we find domains of parameters that give rise to chaos. Wave speeds of traveling wave solutions are also found for certain cases.

Dedication

*I want to dedicate this study to my family and friends for their support throughout my years at
Virginia Tech*

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I want to thank my parents, Davresh and Narmin, and my brother, Armanj, for their constant love and support. My father, Davresh, has had endless patience in teaching me the nature of science through mathematics and physics. The present study is a culmination of the work I have done with my father since my last years as an undergraduate. I sincerely thank the Mathematics Department for allowing me to pursue this project.

I also want to thank my committee members. Dr. John Burns and Dr. Mark Embree have given me unique perspectives and fostered my interest in applied mathematics. I'm grateful to have known my Committee Chair and Advisor Dr. Lizette Zietsman and her husband Dr. Jeff Borggaard; thank you for all your help and guidance over the years.

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Chapter 1

Introduction

A wide range of physical systems are modeled as axially moving strings. Examples include belts, tapes, wires, fibers, etc. These systems have applications in areas like chemical and textile manufacturing. If you assume this string is conductive in an applied magnetic field, the applications broaden. The problem we consider is the motion of a current-carrying string under the influence of a magnetic field, while it is being pulled between two rings.

A potential application includes the space elevator. To ensure structural integrity, the elevator cords are made of conductive carbon nanotubes. These nanotubes (analogous to the string) could be controlled or stabilized by a magnetic field to prevent buckling or failure [5]. In our study, we find the parameters that ensure buckling of the string. Another application is the manufacturing of composite materials. This process involves stitching or placing metallic fibers at high speeds to create composites that are lighter and stronger than homogeneous materials [4, 13].

This study is applicable in plasma or particle physics. Controlling the string in our system is analogous to controlling the motion of a particle or plasma. For example, in Synchrotrons, particle trajectories are controlled by magnetic fields to achieve high speed impacts. Fusion reactors use electromagnetic fields to speed up particles such that they collide (forming a plasma) and release energy. If we consider a small segment of the toroidal accelerators or reactors, individual particles can be represented as strings between two fixed points. The high speed trajectories of particles or plasma can be represented by material points of the string. Therefore, the stability regions and approximate solutions derived in this study can be applied to high speed particle trajectories [10]. Additionally, in Chapter 5 we will see Schrödinger-type differential equations arise.

The analysis we conduct will also contribute to simulation software. The most modern types of software have difficulties reproducing electromagnetic interactions with elastic (conductive) continua.

Thus finding analytical results will be vital for developing numerical models. Although we don't formulate it in this study, a finite element scheme can be developed for commercial applications.

As a preliminary step towards deriving the model, we discuss the derivation and important aspects of the wave equation in Appendix A. The wave equation is a second-order partial differential equation (in space and time) that describes vibration through a medium. Note that presenting the wave equation as a Sturm-Liouville problem is applicable to creating acoustic string instruments.

In general, vibrations of axially moving strings have been studied extensively. Transverse vibrations and the stability of axially moving strings without a current have been investigated in [3, 6, 9, 16, 17]. In 1963, Swope and Ames derived and analyzed a linear model for the oscillations of a moving (being pulled) threadline between two fixed points. This system is applicable in manufacturing; for example, it describes the motion of a string in a sewing machine. They analyze the system in an infinite and a finite setting. The analysis finds the effects of the ratio of winding speed (pulling speed) to wave speed. They find in the limit as winding speed approaches wave speed, there is a growing chance of the threadline breaking or buckling [17].

A large number of studies have also been devoted to the stability of lateral vibrations of current carrying strings and rods. Stability of a straight form of the current carrying bars and strings are investigated by [7, 11, 16]. The first results in this direction were obtained in 1961 by [11], where the stability of a cord of plasma was considered. Leontovich and Shafranov studied a conductive rod in a longitudinal magnetic field. We don't include it in our study, but Leontovich and Shafranov considered the magnetic field generated by deforming the conductor. Through Maxwell's equations and Newton's second law they derive governing equations to calculate the magnetic field and current distribution of the system. Leontovich and Shafranov find critical parameters at which the system buckles or deforms, and they conclude that the system is inherently unstable. In other words, they find the critical value of the magnetic field in a linear setting when the straight line form of the plasma loses stability.

Chattopadhyay and Moon examined the instability of a current carrying rod in a magnetic field. They compute electrostatic potential inside and outside the rod, thus electric field and current distribution is known for the whole space. In the case of nonstationary current flowing through a

string, Chattopadhyay and Moon show that for some values of parameters the string falls into a chaotic state. Furthermore, they confirmed the results experimentally. The analogous problem for the Duffing equation is considered by Sanchez and Nayfeh in [15].

Peter Wolfe studied the equilibrium states of an elastic conductor in a magnetic field. Wolfe used a bifurcation parameter dependent on magnetic field and current. The eigenvalues of Wolfe's linearized problem show where bifurcation branches occur (i.e. nontrivial solutions) [18]. In Appendix D, we present Wolfe's derivation of the equilibrium states. In Chapter 3, we apply a similar analysis to our system and compare it to Wolfe's bifurcation results.

To our knowledge, there are no studies considering an axially moving current-carrying string in a magnetic field. The model derived here is in good comparison with those in [2, 11, 17, 18].

The chapters to follow will cover the aforementioned modeling and analysis of a moving current-carrying string in a magnetic field. Chapter 2 presents a model to describe the motion of the string. Newton's law and momentum balance are used to find a second order (in space and time) nonlinear partial differential equation that governs the motion of the system. The bifurcation and stability of the governing equation is studied in Chapter 3. We consider the static stability in the linear and nonlinear cases, and the dynamic stability in the linear case. In Chapter 4, using Galerkin orthogonalization, the equilibria of the string are categorized. Chapter 5 examines the system with a periodic current. The method of multiple scales [14] is used to approximate solutions for resonance vibrations. The stability of these solutions is also examined. In Chapter 6 traveling-wave type solutions are shown to exist in an infinite setting. The wave speeds are computed for certain cases. We see there is a certain case in which two traveling waves propagate in the same direction.

Chapter 2

Modeling the Equation of Motion

Consider an infinite conductive string in a homogeneous magnetic field $\vec{B}_0 = B_0 \vec{i}_x$, where B_0 is constant. The string is moving with a constant speed $\vec{V} = V_0 \vec{i}_x$ through two rings, which are located a distance l from each other (figure 2.1). The motion of the string is considered in a fixed Cartesian coordinate system (x, y, z) , with unit vectors $\vec{i}_x, \vec{i}_y, \vec{i}_z$. The center of one of the rings is assumed to be the center of the coordinate system. The influence of gravity is neglected. The weight of a unit length of the string is assumed to be constant. We assume the position of each material point of the string is constant. We identify the position of each material point of the string through its arc length $s \in [0, l]$ (s is an arc length from the beginning of the coordinate system to the initial point). In the three dimensional Euclidean space we denote a displacement of the string

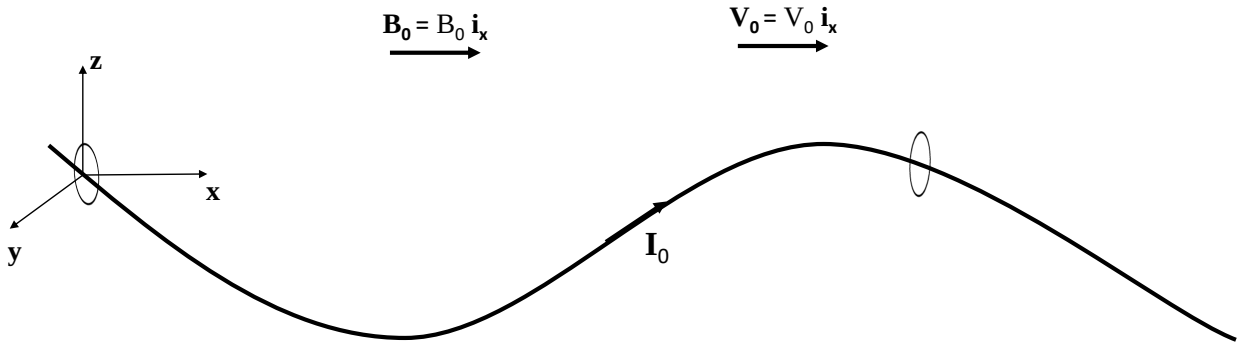


Figure 2.1: Moving current carrying string in a magnetic field $\mathbf{B}_0$

$\vec{r}(s, t) = (x(s, t), y(s, t), z(s, t))$ and assume that $\vec{k} = \frac{\partial x}{\partial s} \vec{i}_x + \frac{\partial y}{\partial s} \vec{i}_y + \frac{\partial z}{\partial s} \vec{i}_z = \frac{\partial \vec{r}}{\partial s}$ is a unit vector directed along the tangent to the deformed line of the string $\left(|\vec{k}| = \sqrt{\left(\frac{\partial x}{\partial s}\right)^2 + \left(\frac{\partial y}{\partial s}\right)^2 + \left(\frac{\partial z}{\partial s}\right)^2}\right)$. If we assume a current $\vec{I} = I_0(t) \vec{k}$ flows through the string, then the magnetic field will influence the system. Laplace force, a sub-case of Lorentz force, is the force that causes deflection of a current carrying wire in a magnetic field. The force acting on string is given by

$$\vec{F} = I_0 B_0 (\vec{k} \times \vec{i}_x). \quad (2.1)$$

The motion of the string is described by the system of equations

$$\frac{\partial \vec{Q}}{\partial s} + \vec{F} = \rho \vec{R}, \quad \vec{Q} \times \vec{k} = 0, \quad \left(\frac{\partial \vec{r}}{\partial s}\right) = 1, \quad (2.2)$$

where $\vec{Q}$ is the vector of internal stresses, ρ is the mass per unit length and the vector $\vec{R}$ has the following projections on the coordinate axis:

$$R_x = 0, \quad R_y = \frac{\partial^2 y}{\partial t^2} + 2V_0 \frac{\partial^2 y}{\partial s \partial t} + V_0^2 \frac{\partial^2 y}{\partial s^2} + \eta \frac{\partial y}{\partial t}, \quad (2.3)$$

$$R_z = \frac{\partial^2 z}{\partial t^2} + 2V_0 \frac{\partial^2 z}{\partial s \partial t} + V_0^2 \frac{\partial^2 z}{\partial s^2} + \eta \frac{\partial z}{\partial t}$$

where η is the coefficient of external damping (viscosity) [3, 7, 11, 12, 16, 17]. The vector $\vec{R}$ describes inertial forces. Notice that equation (2.2) is written in a moving coordinate system, thus we apply the change of coordinates $s \rightarrow s + V_0 t$ and $\frac{\partial}{\partial t} \rightarrow \frac{\partial}{\partial t} + V_0 \frac{\partial}{\partial s}$. This follows the derivation by Swope and Ames in [17]. Essentially, the inertial forces in the y and z direction, i.e. R_y and R_z , are described by the threadline equation derived by Swope and Ames. Since the string is being pulled in the axial direction, we assume inertial force in the x -direction does not have a contribution.

From equation (2.2), the projection of $\frac{\partial \vec{Q}}{\partial s} + \vec{F} = \rho \vec{R}$ on the coordinate axes is

$$\begin{aligned} \frac{\partial Q_x}{\partial s} &= 0, \\ \frac{\partial Q_y}{\partial s} + I_0 B_0 \frac{\partial z}{\partial s} &= \rho \left(\frac{\partial^2 y}{\partial t^2} + 2V_0 \frac{\partial^2 y}{\partial s \partial t} + V_0^2 \frac{\partial^2 y}{\partial s^2} + \eta \frac{\partial y}{\partial t} \right), \\ \frac{\partial Q_z}{\partial s} - I_0 B_0 \frac{\partial y}{\partial s} &= \rho \left(\frac{\partial^2 z}{\partial t^2} + 2V_0 \frac{\partial^2 z}{\partial s \partial t} + V_0^2 \frac{\partial^2 z}{\partial s^2} + \eta \frac{\partial z}{\partial t} \right), \end{aligned} \quad (2.4)$$

the projection of the momentum balance $\vec{Q} \times \vec{k} = 0$ and the arc-length are

$$Q_y \frac{\partial x}{\partial s} = Q_x \frac{\partial y}{\partial s}, \quad Q_z \frac{\partial x}{\partial s} = Q_x \frac{\partial z}{\partial s}, \quad Q_y \frac{\partial z}{\partial s} = Q_z \frac{\partial y}{\partial s}, \quad \text{and} \quad (2.5)$$

$$\left(\frac{\partial x}{\partial s}\right)^2 + \left(\frac{\partial y}{\partial s}\right)^2 + \left(\frac{\partial z}{\partial s}\right)^2 = 1.$$

Determining Q_y, Q_z , and $\frac{\partial x}{\partial s}$ from (2.5) and using (2.4) we get

$$\frac{\partial}{\partial s} \left\{ Q_x \frac{\partial y}{\partial s} \left[1 - \left(\frac{\partial y}{\partial s}\right)^2 - \left(\frac{\partial z}{\partial s}\right)^2 \right]^{-1/2} + I_0 B_0 z \right\} = \rho \left(\frac{\partial^2 y}{\partial t^2} + 2V_0 \frac{\partial^2 y}{\partial s \partial t} + V_0^2 \frac{\partial^2 y}{\partial s^2} + \eta \frac{\partial y}{\partial t} \right), \quad (2.6)$$

$$\frac{\partial}{\partial s} \left\{ Q_x \frac{\partial z}{\partial s} \left[1 - \left(\frac{\partial y}{\partial s}\right)^2 - \left(\frac{\partial z}{\partial s}\right)^2 \right]^{-1/2} - I_0 B_0 y \right\} = \rho \left(\frac{\partial^2 z}{\partial t^2} + 2V_0 \frac{\partial^2 z}{\partial s \partial t} + V_0^2 \frac{\partial^2 z}{\partial s^2} + \eta \frac{\partial z}{\partial t} \right), \quad (2.7)$$

$$\frac{\partial Q_x}{\partial s} = 0. \quad (2.8)$$

From equation (2.8) we have $Q_x(x, y, z, t) = P_0$, a constant. Using the complex function $u(s, t) = y(s, t) + iz(s, t)$ in (2.6)-(2.8) we obtain the governing equation for lateral vibrations of the string:

$$\rho \left[\frac{\partial^2 u(s, t)}{\partial t^2} + \eta \frac{\partial u(s, t)}{\partial t} + 2V_0 \frac{\partial^2 u(s, t)}{\partial s \partial t} + V_0^2 \frac{\partial^2 u(s, t)}{\partial s^2} \right] = \frac{\partial}{\partial s} \left(\frac{P_0 \frac{\partial u(s, t)}{\partial s}}{\sqrt{1 - \left| \frac{\partial u(s, t)}{\partial s} \right|^2}} \right) - i I_0 B_0 \frac{\partial u(s, t)}{\partial s}, \quad (2.9)$$

where $0 < s < l$. Equation (2.9) is considered with the boundary conditions

$$u(0, t) = u(l, t) = 0. \quad (2.10)$$

These boundary conditions specify that the string is fixed at the rings or endpoints. Initial conditions will be assigned appropriately. Notice that there were no limitations imposed on I_0 in the derivation of equation (2.9). This suggests we can have a time-varying current (i.e. $I_0 = I_0(t)$), which we will revisit in Chapter 5. If we assume that the deformations are negligible in magnitude to the entire length of the string, i.e. $s \simeq x$ and $1 - \left| \frac{\partial u}{\partial s} \right|^2 \simeq 1$, equation (2.9) becomes linear:

$$\frac{\rho}{P_0} \frac{\partial^2 u(s, t)}{\partial t^2} + \frac{\rho \eta}{P_0} \frac{\partial u(s, t)}{\partial t} + \frac{2\rho V_0}{P_0} \frac{\partial^2 u(s, t)}{\partial s \partial t} = \left(1 - \frac{\rho V_0^2}{P_0} \right) \frac{\partial^2 u(s, t)}{\partial s^2} - i \frac{I_0 B_0}{P_0} \frac{\partial u(s, t)}{\partial s}. \quad (2.11)$$

Notice that the governing equation (2.9) is a second order nonlinear partial differential equation, where the solution $u(s, t)$ is described in terms of $y(s, t)$ and $z(s, t)$. Finding the solution to (2.9) amounts to finding the solutions $y(s, t)$ and $z(s, t)$ of two coupled second order nonlinear partial differential equations. If we replaced the string in our system with a rod, the cross-product in the momentum equation would equate to a momentum dependent term. In this case, the system would become a fourth order nonlinear partial differential equation, i.e. with this assumption, our system is similar to the beam equation.

The governing equation (2.9) is a generalization of previously mentioned studies. Neglecting the magnetic field, viscosity, and current reduces equation (2.9) to

$$\frac{\partial^2 u}{\partial t^2} + \bar{\alpha} \frac{\partial^2 u}{\partial s \partial t} + \bar{\beta} \frac{\partial^2 u}{\partial s^2} = 0,$$

where $\bar{\alpha} = 2V_0$ and $\bar{\beta} = V_0^2$. Note that this is the threadline equation analyzed by Swope and Ames in [17]. Similar to their study, we will find the wave-speeds of the generalized equation (2.9) in an infinite setting.

The derivation of (2.9) and (2.11) assume that gravity and the magnetic fields generated from the deformation are negligible compared to the applied forces. Also, the magnetic field is assumed to be given, thus we avoid calculating the electric and magnetic fields through Maxwell's equations. Note that to generate the magnetic field, $\vec{B}_0$, we can surround the string with N currents. A possible configuration is shown in figure 2.2. Suppose I_k denotes the current running through the k th current. Let (y_k, z_k) denote the position of the currents, with the string positioned at $(0, 0)$. The magnetic field applied on the string surrounded by N currents is given by

$$\vec{B}_0 = \sum_{k=1}^N I_k \frac{(y - y_k)\vec{i}_y + (z - z_k)\vec{i}_z}{\sqrt{(y - y_k)^2 + (z - z_k)^2}}. \quad (2.12)$$

This is obtained by applying Maxwell's equations to the system with N distributed currents (Figure 2.2).

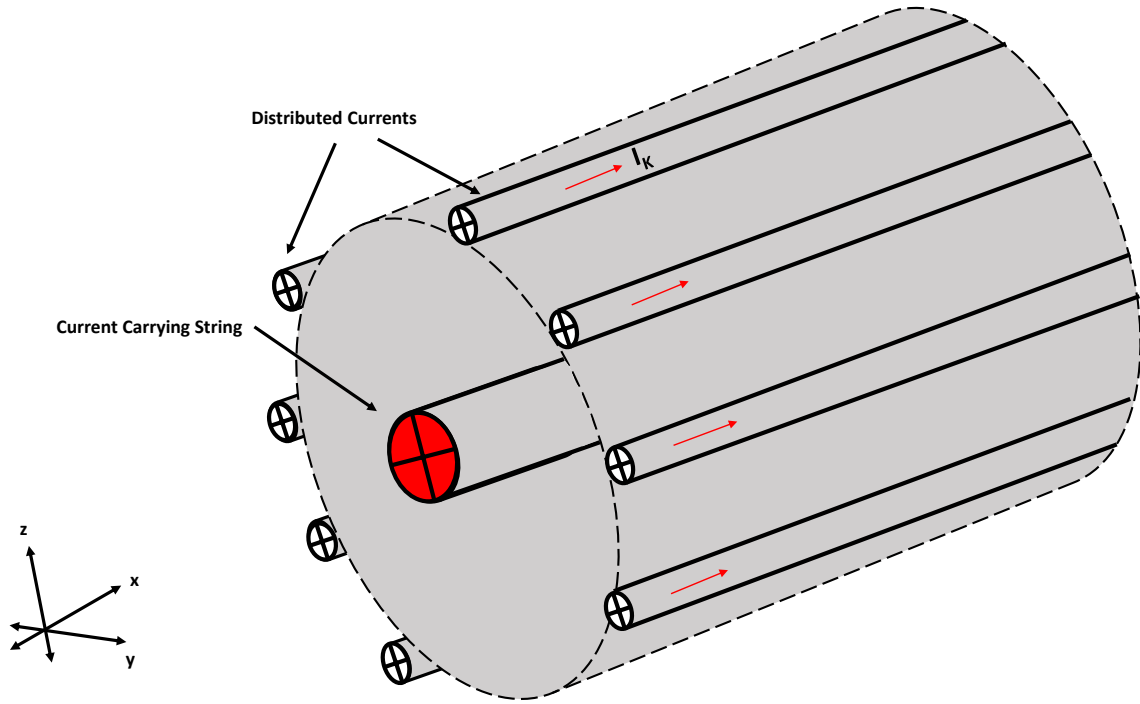
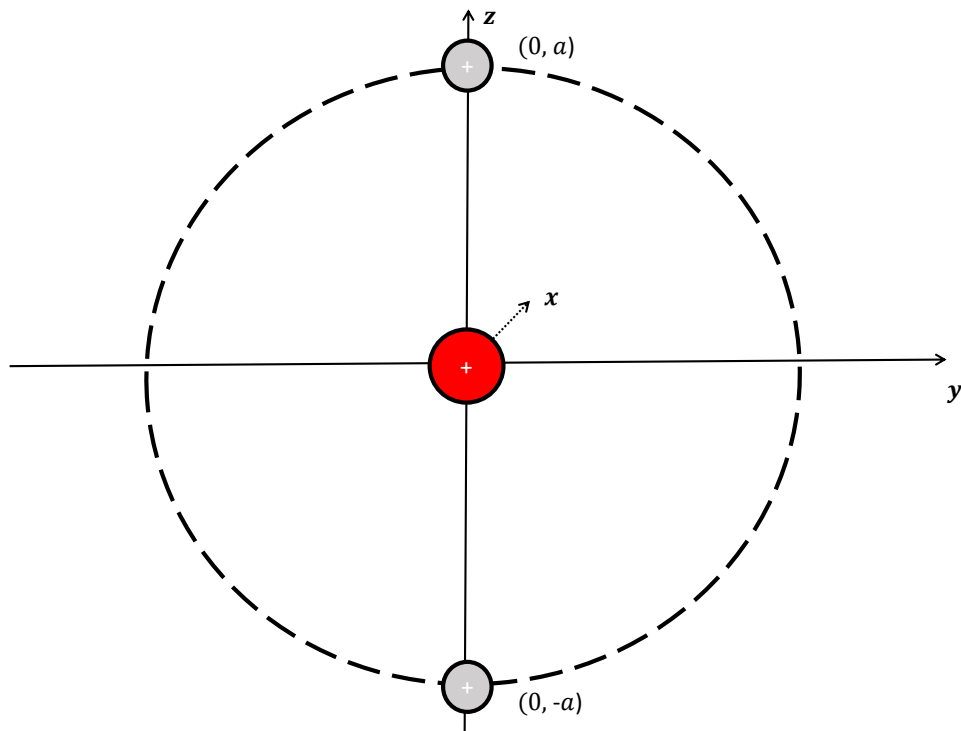


Figure 2.2: Currents distributed around the string to generate magnetic field $\vec{B}_0$.

For example, the system with two currents shown in figure (2.3) has the following magnetic field

$$\vec{B}_{\pm} = 2I_1 \frac{(z \mp a)\vec{i}_z - y\vec{i}_y}{\sqrt{(z \mp a)^2 + y^2}}. \quad (2.13)$$

The magnetic field in equation (2.12) is applied in the y - and z -directions. The orientation of the currents can be chosen such that the magnetic field is only applied in the x -direction. For example, currents placed perpendicular to the string (like a coil) will generate the desired magnetic field, $\vec{B}_0 = B_0\vec{i}_x$.



1

Figure 2.3: Two currents distributed around the string to generate magnetic field $\vec{B}_0$.

Chapter 3

Stability of the String

The governing equation will be analyzed a static and dynamic setting. We assume that the string is fixed at two points a distance ℓ apart (boundary condition (2.10)). In the static case, time is fixed (i.e. $\frac{\partial u}{\partial t} = 0$). This allows us to find equilibrium solutions, describing the possible forms the string can take at any given time. The analysis is similar to finding buckling nodes of a load-bearing beam. We will consider both the linear (2.11) and nonlinear (2.9) equations in the static case. Recall in the linear case we are assuming the magnitude of perturbations are negligible to the size of the string. In the linear static case, we follow and compare our results with Wolfe's approach shown in Appendix B and [18]. This will allow us to find parameters for which the string goes from a straight trivial shape to a deformed spiral shape. Similarly, we find the parameter for which the string buckles in the nonlinear static case. Both equations in the static case lead to Sturm-Liouville eigenvalue problems with boundary conditions (2.10).

The dynamic case considers the string's motion through time. We only consider the linearized equation (2.11) in the dynamic setting. Our notion of stability is defined for this case and a bifurcation of solutions is described.

We show that the condition on the parameters for straight trivial solutions in the linear static case diverge from the conditions in the linear dynamic case. The conditions for each case are plotted in section 3.2. The divergence suggests that the system is nonconservative, i.e., energy is dissipated through time. This result is also seen in Appendix C, where we show uniqueness of solutions and that energy is bounded for a particular case of our string.

3.1 Static Linear and Nonlinear Equilibrium States

Recall equations (2.9) and (2.10):

$$\rho \left[\frac{\partial^2 u(s, t)}{\partial t^2} + \eta \frac{\partial u(s, t)}{\partial t} + 2V_0 \frac{\partial^2 u(s, t)}{\partial s \partial t} + V_0^2 \frac{\partial^2 u(s, t)}{\partial s^2} \right] = \frac{\partial}{\partial s} \left(\frac{P_0 \frac{\partial u(s, t)}{\partial s}}{\sqrt{1 - \left| \frac{\partial u(s, t)}{\partial s} \right|^2}} \right) - iI_0 B_0 \frac{\partial u(s, t)}{\partial s},$$

and

$$\frac{\rho}{P_0} \frac{\partial^2 u(s, t)}{\partial t^2} + \frac{\rho \eta}{P_0} \frac{\partial u(s, t)}{\partial t} + \frac{2\rho V_0}{P_0} \frac{\partial^2 u(s, t)}{\partial s \partial t} = \left(1 - \frac{\rho V_0^2}{P_0} \right) \frac{\partial^2 u(s, t)}{\partial s^2} - i \frac{I_0 B_0}{P_0} \frac{\partial u(s, t)}{\partial s}.$$

Consider the nonlinear equation (2.9) with boundary conditions (2.10) for the equilibrium or static states (i.e. $\frac{\partial u}{\partial t} = 0$). Physically, this is looking at the string for a fixed instance in time. Depending on critical parameter values, the straight-line form of the string will deform and assume a spiral configuration (buckle). Parameter regions corresponding to deformations in the string will be defined as regions of instability. We are in essence looking at the Sturm-Liouville eigenvalue problem

$$\begin{cases} \rho V_0^2 \frac{d^2 u(s)}{ds^2} - \frac{d}{ds} \left(\frac{P_0 \frac{du(s)}{ds}}{\sqrt{1 - \left| \frac{du(s)}{ds} \right|^2}} \right) + iI_0 B_0 \frac{du(s)}{ds} = 0, & \text{where } 0 < s < \ell, \\ u(0) = u(\ell) = 0. \end{cases}$$

A possible solution of the static nonlinear equation above can have the form

$$u(s) = A + B e^{i\gamma s}, \quad (3.1)$$

where A and B are constants, and the parameter γ satisfies the condition

$$\frac{P_0 \gamma}{\sqrt{1 - \gamma^2 |A|^2}} = \rho V_0^2 \gamma + I_0 B_0. \quad (3.2)$$

Applying the boundary conditions above results in $A = -B$ and $\gamma = \lambda_k = \frac{2\pi k}{\ell}$, $k = \pm 1, \pm 2, \dots$

If the parameters of equation (2.9) satisfy condition (3.2), then equations (2.9)-(2.10) will have nontrivial solutions, $u(s)$. In this case, the string has a configuration (nontrivial solution) that is a superposition of the basis functions, $\phi_k(x)$, $k = 1, 2, \dots$, corresponding to eigenvalues, λ_k . Physically this means if parameters satisfy (3.2), then the string enters a buckling regime. Buckling is a result of stresses caused by compression. This implies that the magnetic field, current, and axial pulling speed can be selected such that buckling is avoided. Returning to the discussion of particle accelerators and fusion reactors, note that parameters preserving the straight-line form of the string will be vital for stability in these systems.

Now consider the string is in a fixed time (static) with the assumption that the magnitude of perturbation is negligible compared to the length of the entire string (linear). The static linearized problem is given by

$$\begin{cases} \left(1 - \frac{\rho V_0^2}{P_0}\right) \frac{d^2 u(s)}{ds^2} - i \frac{I_0 B_0}{P_0} \frac{du(s)}{ds} = 0, & \text{where } 0 < s < \ell, \\ u(0) = u(\ell) = 0. \end{cases}$$

This is a generalization of Wolfe's linear study in [18], since we consider axial motion. We discuss key results from [18] in Appendix B. Note that the static problem is analogous to the system (B.3), (B.4). However, Wolfe's boundary conditions only fixed the left endpoint but allowed the right to be arbitrary. This leads to branches of nontrivial solutions (bifurcations).

Similar to (3.2), assuming the solution form (3.1) in the linear static case gives the buckling condition

$$\frac{\rho V_0^2}{P_0} + \frac{I_0 B_0}{P_0 \gamma} = 1, \quad (3.3)$$

which can also be obtained from (3.2) by assuming $1 - \gamma^2 |A|^2 \approx 1$. Using the boundary conditions, we have $\gamma = \lambda_k = \frac{2\pi k}{\ell}$, for $k = \pm 1, \pm 2, \dots$. Equation (3.3) can be solved for the eigenvalue (or bifurcation parameter), which gives

$$\gamma = \frac{I_0 B_0 / P_0}{1 - \rho V_0^2 / P_0}. \quad (3.4)$$

Note in [18], the system was dimensionless and neglected axial motion, the eigenvalue was $\lambda = IB$. Every solution in the linear static case can be generated by basis functions corresponding to the

eigenvalues λ_k . The condition for straight, unbuckled solutions in the linear static case is

$$\frac{\rho V_0^2}{P_0} + \frac{I_0 B_0}{P_0 \lambda_k} < 1. \quad (3.5)$$

When parameters are chosen outside of the region described in (3.5), the string will undergo a deformation. The shape of the string will depend on the structure of the eigenfunctions $\phi_k(s)$ corresponding to the eigenvalues λ_k . The parametric representation of the basis functions corresponding to these eigenvalues are shown below. In figures 3.1–3.3, let $s = x_k(s)$, $\text{Re}\phi_k(s) = 1 - \cos(\lambda_k s) = y_k(s)$, and $\text{Im}\phi_k(s) = -\sin(\lambda_k s) = z_k(s)$, for $k = 1, 2, 3$. When (3.5) is satisfied, we have condition

$$0 < \frac{I_0 B_0 / P_0}{1 - \rho V_0^2 / P_0} < \frac{2\pi}{\ell}.$$

Parameters satisfying this condition will result in trivial solutions, i.e., the string will remain straight. For parameters outside this interval, the string will deform.

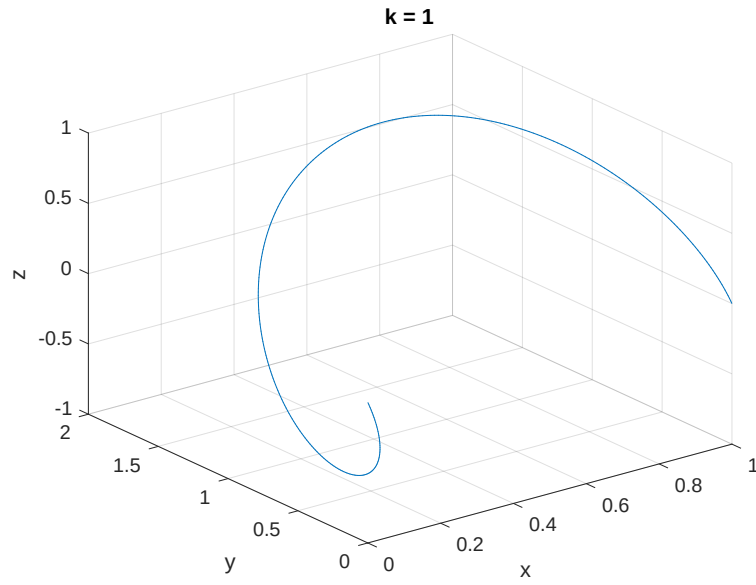


Figure 3.1: $\phi_k(s)$ plotted parametrically for $k = 1$ with $\ell = 1$. In this case, the parameter relationship or eigenvalues satisfy $2\pi < \frac{I_0 B_0 / P_0}{1 - \rho V_0^2 / P_0} < 4\pi$. Physically this means for parameters in the aforementioned interval, the string will have a kink or one spiral corresponding to $\phi_1(s)$.

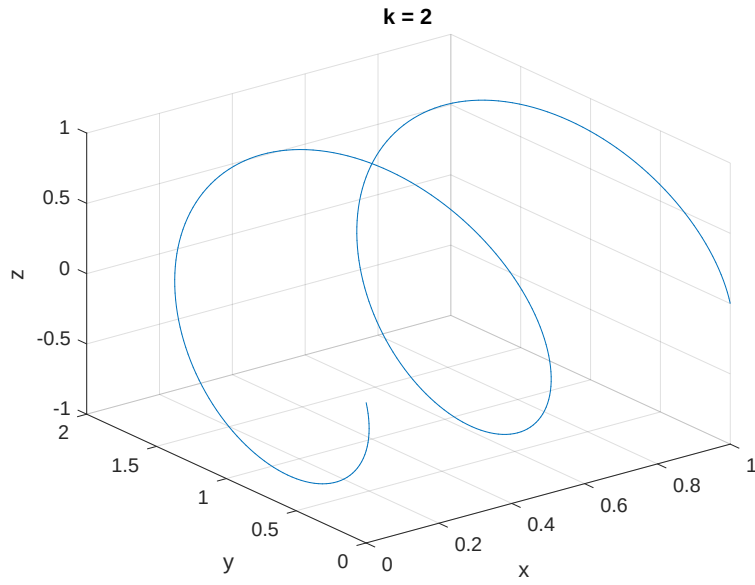


Figure 3.2: $\phi_k(s)$ plotted parametrically for $k = 2$ with $\ell = 1$. In this case, the parameter relationship or eigenvalues satisfy $4\pi < \frac{I_0 B_0 / P_0}{1 - \rho V_0^2 / P_0} < 6\pi$. Physically this means for parameters in the aforementioned interval, the string will have two kinks or spirals corresponding to $\phi_2(s)$.

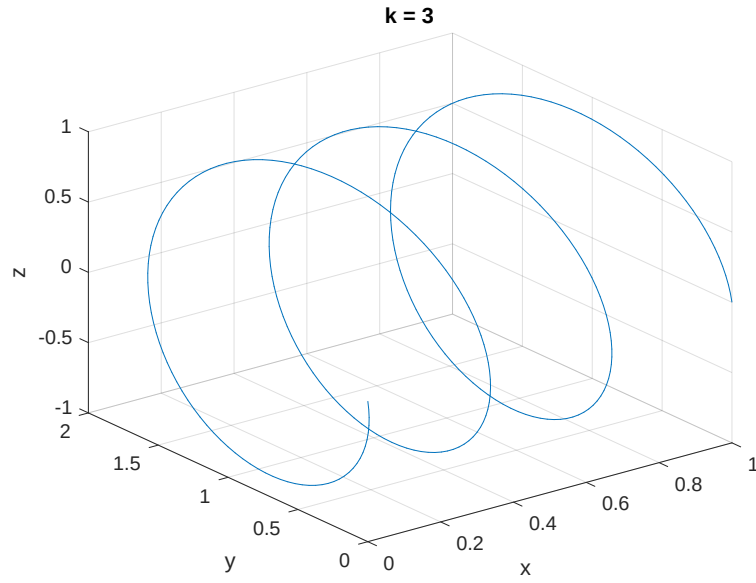


Figure 3.3: $\phi_k(s)$ plotted parametrically for $k = 3$ with $\ell = 1$. In this case, the parameter relationship or eigenvalues satisfy $6\pi < \frac{I_0 B_0 / P_0}{1 - \rho V_0^2 / P_0} < 8\pi$. Physically this means for parameters in the aforementioned interval, the string will have three kinks or spirals corresponding to $\phi_3(s)$.

For the linear static case, if we assume no axial motion, $V_0 = 0$, one arrives at the case studied in [18]. As previously mentioned, the main difference is that our study assumes fixed boundary conditions (we let $b = 0$ in boundary condition (B.4)). This results in a discrete number of bifurcation points at which the string is deformed (i.e., $\lambda_k = 2\pi k/\ell$, $k = 1, 2, \dots$). In [18], Wolfe assumes that the right boundary point is fixed at an arbitrary point ($b \geq 1$), which leads to branches at which nontrivial (deformed) solutions occur. Our results are compared with [18] in table 3.1, where $\hat{N}(b)$ describes the tension at a given point. The remaining results and notation for Wolfe's system are shown in Appendix B.

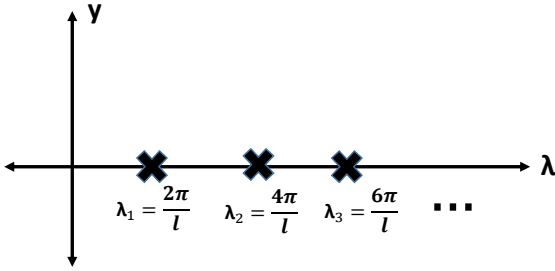
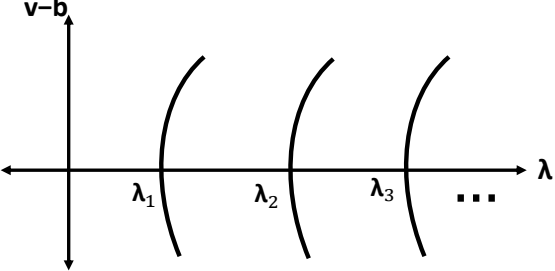
	Our System	Wolfe's System
Assumptions	Current carrying, axially moving string in a magnetic field in a linear static setting. Boundary conditions are $u(0)=u(l)=0$.	Dimensionless current carrying string in a magnetic field in a linear static setting. Boundary Conditions are $u(0)=0$, $u(l)=b$.
Eigenvalues	$\Upsilon = \frac{I_0 B_0 / P_0}{1 - \rho V_0 / P_0}, \quad \lambda_k = \frac{2\pi k}{l}$	$\lambda = IB, \quad \lambda_m = \frac{2\pi m \hat{N}(b)}{b}$
Bifurcation of solutions	 Note: We get discrete points of bifurcation since we fix both boundary conditions.	 Note: Wolfe's case considers an arbitrary right boundary point. Thus the bifurcation diagram gives regions of bifurcation.

Table 3.1: Comparison with Wolfe's results in [18]

3.2 Dynamic Linear Stability

In general, it is difficult to determine a condition for dynamic stability of the nonlinear problem (2.9)–(2.10). Dynamic instability is associated with an unbounded increase in the amplitude of the vibrations. We derive a condition for dynamic stability of the linearized problem (2.10)–(2.11), given by

$$\begin{cases} \frac{\rho}{P_0} \frac{\partial^2 u(s, t)}{\partial t^2} + \frac{\rho\eta}{P_0} \frac{\partial u(s, t)}{\partial t} + \frac{2\rho V_0}{P_0} \frac{\partial^2 u(s, t)}{\partial s \partial t} = \left(1 - \frac{\rho V_0^2}{P_0}\right) \frac{\partial^2 u(s, t)}{\partial s^2} - i \frac{I_0 B_0}{P_0} \frac{\partial u(s, t)}{\partial s}, \\ u(0, t) = u(l, t) = 0, \end{cases}$$

where $0 < s < \ell$. The stability of this case is defined as follows; let $\tilde{E}(t)$ be defined as in (C.12),

Definition We say that a solution $u(s, t)$ is **stable** if given any $C > 0$, there exists a $\delta > 0$ such that if $\tilde{E}(0) < \delta$, then $\tilde{E}(t) < C$ for all $t > 0$. Otherwise, the solution $u(s, t)$ is called **unstable**.

Initially bounded solutions can also be regarded as having limited energy, as discussed in Appendix C. The energy argument amounts to having solutions satisfying boundedness in the L_2 -space.

Consider solutions for (2.10)–(2.11) of the form

$$u(s, t) = A e^{\gamma_1 s + i\omega t} + B e^{\gamma_2 s + i\omega t}, \quad (3.6)$$

where A and B are constants, $a = \frac{\rho}{P_0}$, $b = 2aV_0$, $c = 1 - aV_0^2$, $\lambda = \frac{I_0 B_0}{2P_0}$, and

$$\gamma_{1,2} = \frac{i(\lambda + b\omega) \pm \sqrt{-(\lambda + b\omega)^2 - 4ac\omega^2}}{2c}.$$

The factors $e^{\gamma_1 s}$ and $e^{\gamma_2 s}$ are limited or bounded in the interval $0 \leq s \leq \ell$ as $t \rightarrow \infty$. This implies that the stability of $u(s, t)$ depends on ω . In this context, ω is a complex number, i.e., $\omega = \omega_R + i\omega_I$, where ω_R and ω_I are real. Thus we have $e^{i\omega t} = e^{i\omega_R t} e^{-\omega_I t}$. Since $|e^{i\omega_R t}| = 1 < \infty$, we conclude that the stability must depend on ω_I as $t \rightarrow \infty$. This suggests three distinct cases can occur. If $\omega_I > 0$, then $e^{-\omega_I t} \rightarrow 0$ as $t \rightarrow \infty$, and the solution will be stable. If $\omega_I < 0$, then $e^{-\omega_I t} \rightarrow \infty$ as $t \rightarrow \infty$, and the solution will be unstable. If $\omega_I = 0$, then $e^{-\omega_I t} = 1$ as $t \rightarrow \infty$. The solution in this case will give a limit cycle, so the solution can be stable or unstable. Determination of stability in this case requires an additional investigation of the nonlinear terms in equation (2.9).

Using nontrivial solutions of the form (3.6) in (2.10)–(2.11), results in a linear system. The dimensionless determinant of that system is given by

$$4\Omega^2 + 4\beta\sqrt{\alpha}\Omega + \beta^2 - (1 - \alpha)^2 = 0, \quad (3.7)$$

where $\Omega = \omega/\Omega_0$, $\Omega_0 = \lambda_k \sqrt{1/a}$, $\beta = 2\lambda/\lambda_k$, $\lambda_k = 2\pi k/l$, and $\alpha = aV_0^2$. In (3.7) we assume $\beta \geq 0$

and $0 \leq \alpha \leq 1$. Using equation (3.7), leads to the condition for the dynamic stability

$$\left(\frac{I_0 B_0}{\lambda_k P_0} \right)^2 + \frac{\rho V_0^2}{P_0} < 1. \quad (3.8)$$

Notice that conditions (3.5) and (3.8) for linear static and dynamic stability are different (shown in figures 3.4 and 3.5, respectively). This divergence between the linear and nonlinear stability regions suggests that this system is nonconservative. Consequently, the energy is not preserved. This observation is expected since a dissipative force (viscosity) influences the system over time. Furthermore, the magnetic field can act as a damping force. Such observations are considered in Appendix C.

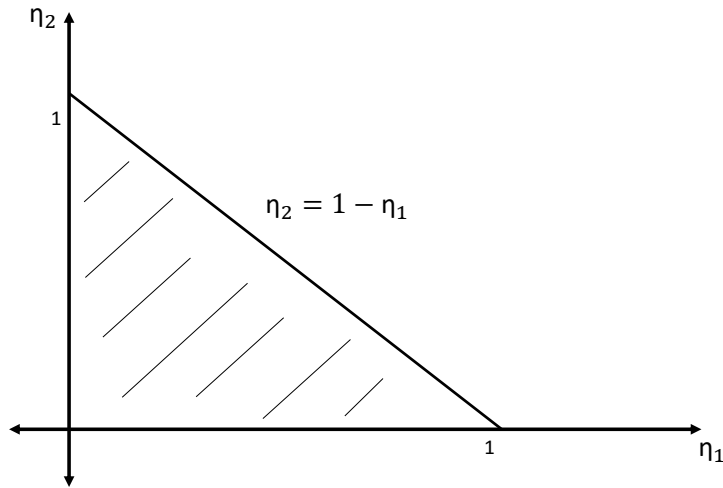


Figure 3.4: The stability region of the static linear equation. The condition is given by equation (3.5), where, $\frac{\rho V_0^2}{P_0} = \eta_1$ and $\frac{I_0 B_0}{P_0 \lambda_k} = \eta_2$.

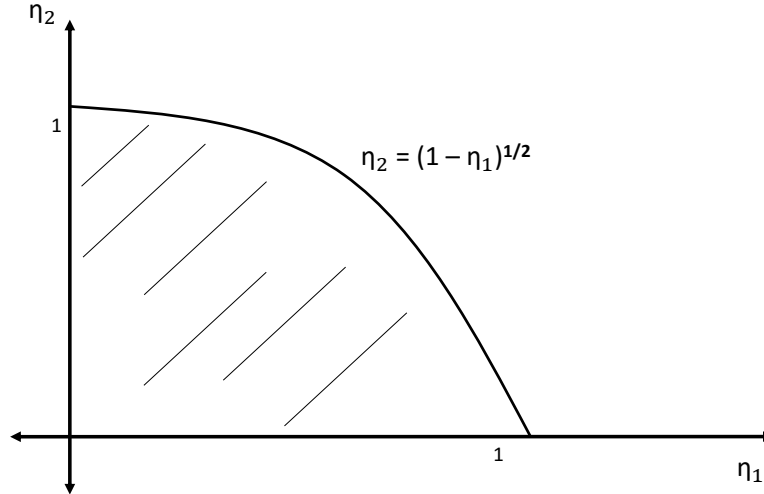


Figure 3.5: Stability region of the dynamic linear equation. The condition is given by equation (3.8), where, $\frac{\rho V_0^2}{P_0} = \eta_1$ and $\frac{I_0 B_0}{P_0 \lambda_k} = \eta_2$.

Returning to the discussion of dynamic stability, notice in equation (3.7), $\text{Im}(\Omega)$ is analogous to ω_I . Thus $\text{Im}(\Omega)$ will give an indication of the system's stability. Equation (3.7) has two complex roots:

$$\Omega_{1,2}(\alpha, \beta) = \frac{-\beta\sqrt{\alpha} \pm \sqrt{\beta^2\alpha - \beta^2 + (1 - \alpha)^2}}{2}. \quad (3.9)$$

If the discriminant, $D(\alpha, \beta) = \beta^2\alpha - \beta^2 + (1 - \alpha)^2$, is positive, the roots Ω_1 and Ω_2 are real and the solution $u(s, t)$ is periodic. As previously mentioned, stability in the periodic case can be investigated by analyzing the nonlinear terms in equation (2.9).

To arrive at a general result for the roots of equation (3.7), we consider specific cases of (3.9). Suppose $\alpha = 0$ and β is increased from 0 to ∞ . The solutions to equation (3.7) will be given by

$$\Omega_I(0, \beta) = \frac{-\sqrt{1 - \beta^2}}{2} \quad \text{and} \quad \Omega_{II}(0, \beta) = \frac{\sqrt{1 - \beta^2}}{2}. \quad (3.10)$$

We can investigate the behavior of the roots as $\beta \rightarrow \infty$. The approximate trajectories of the roots of equation (3.7) in this case are shown in figure (3.6). For $0 < \beta < 1$, the roots are real numbers, but for $\beta > 1$ the roots lie on the $\text{Im}(\Omega)$ axis.

As an example, consider the case for $\alpha = 1/9$ and $\beta \rightarrow \infty$. The approximate root trajectories

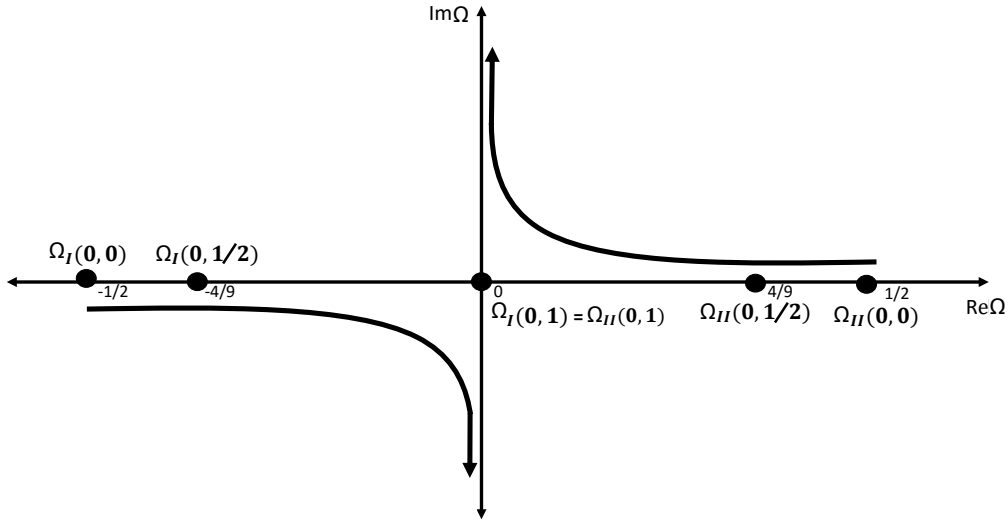


Figure 3.6: Approximate trajectory of the roots given in equation (3.10) with $\alpha = 0$ and $\beta \rightarrow \infty$. The arrows on the trajectories indicate the direction of the roots as β is increased.

are shown in figure 3.7. In this case, the point at which the branches of solutions split is shifted. Furthermore, after leaving the real axis, the roots do not approach the imaginary axis, as in the previous case.

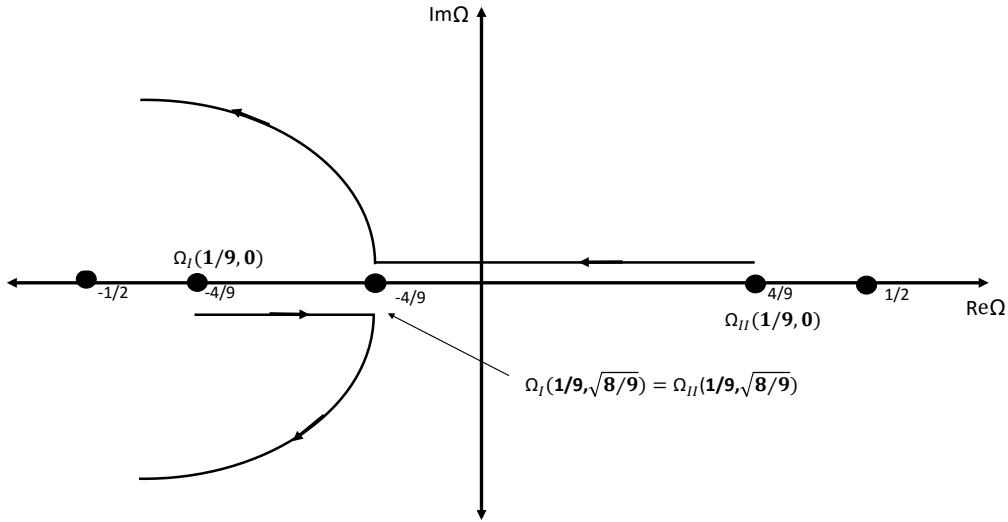


Figure 3.7: Approximate trajectory of the roots given in equation (3.10) with $\alpha = 1/9$ and $\beta \rightarrow \infty$. The arrows on the trajectories indicate the direction of the roots as β is increased.

In figure 3.8, the approximate trajectories of the roots of equation (3.7) are shown in the complex plane Ω , for a range of values where $0 \leq \alpha \leq 0.25$ and β increasing from 0 to ∞ . When $\beta = 0$, the motion of the roots start from points A and A_1 . Points B and B_1 correspond to $\beta = \beta_0 = \sqrt{\alpha(1 - \alpha)}$,

points C and C_1 correspond to $\beta = \beta_{0S} = 1 - \alpha$, and point D corresponds to $\beta = \beta_{0D} = \sqrt{1 - \alpha}$. When the parameter β increases from 0 to $\beta = \beta_{0D}$ the roots of equation (3.7) move through the $\text{Re}(\Omega)$ axis, describing stable solutions for the motion of the string. Starting from point D , where $\beta = \beta_{0D}$ the roots of equation (3.7) come out to the complex plane and the vibration of the string increases with time. As the imaginary components of the roots increase the string loses stability. The value $\beta = \beta_{0S}$ corresponds to static instability, which coincides with equation (3.5). As we can see $\beta_{0S} < \beta_{0D}$, so the static and dynamic problems give different ranges for stability of the string. Here, stability is defined in the sense that the string is in a straight-line form, or attains a straight-line form over time. Similar behavior of roots can be observed for $0.25 \leq \alpha \leq 0.5$. In this case, $\Omega_I(\alpha, \beta_0) < \Omega_I(\alpha, \beta_{0C}) < \Omega_I(\alpha, 0) < \Omega_I(\alpha, \beta_{0D})$, and for $0.5 \leq \alpha \leq 1$ it is necessary to take into account that $\Omega_I(\alpha, \beta_0) < \Omega_I(\alpha, \beta_{0C}) < \Omega_I(\alpha, \beta_{0D}) < \Omega_I(\alpha, 0)$.

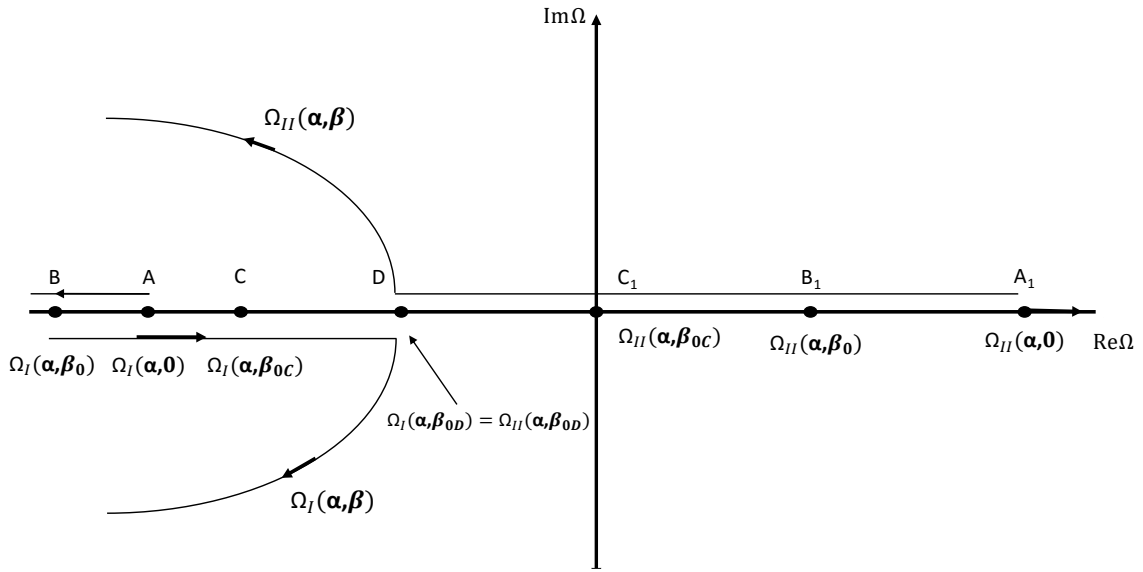


Figure 3.8: Approximate trajectory of the roots of equation (3.7) for $0 \leq \alpha \leq 0.25$ and $0 \leq \beta < \infty$.

In summary, we have found the conditions for which the straight-line form of the string buckles in the dynamic linear case. The condition for dynamic stability is described in (3.8).

Chapter 4

Investigation of Equilibrium States

The general nonlinear equation (2.9) with boundary conditions (2.10) is difficult to study directly. So we apply Galerkin Orthogonalization process to equation (2.9) and consider one mode approximations of the solution. We represent the solution of (2.9) in the form

$$u(s, t) = \sum_{n=0}^{\infty} A_n(t) \Phi_n(s), \quad (4.1)$$

where $\Phi_n(s) = e^{i\lambda_n s}$, $\lambda_n = 2\pi n/l$ ($n = 0, 1, \dots$). From the boundary conditions (2.10) we have $\sum_{n=0}^{\infty} A_n(t) = 0$. We introduce the following scalar product $\langle a, b \rangle = \frac{1}{l} \int_0^l a(t) \overline{b(t)} dt$, where the function $\overline{b(t)}$ is the complex conjugate of $b(t)$. Later we will keep two modes of (4.1) (the members with indices $n = 0$ and $n = m$). Substituting Equation (4.1) into (2.9) and applying Galerkin Orthogonalization procedure we obtain the following ordinary nonlinear differential equation for the amplitude of vibration of the string:

$$\frac{d^2 A_n(t)}{dt^2} + (\eta + 2iV_0\lambda_n) \frac{dA_n(t)}{dt} - V_0^2 A_n(t) \lambda_n^2 - \frac{I_0 B_0 \lambda_n A_n(t)}{\rho} + \frac{P_0 A_n(t) \lambda_n^2}{\rho \sqrt{1 - |A_n(t)|^2 \lambda_n^2}} = 0. \quad (4.2)$$

After denoting $t = \frac{\tau}{\Omega_0}$ and taking into account that $\frac{P_0 \lambda_n^2}{\rho} = \Omega_0^2$ equation (4.2) will take the form

$$\frac{d^2 A_n}{d\tau^2} + \left(\frac{\eta}{\Omega_0} + \frac{2iV_0\lambda_n}{\Omega_0} \right) \frac{dA_n}{d\tau} - \left(\frac{V_0^2 \rho}{P_0} + \frac{I_0 B_0}{\lambda_n P_0} \right) A_n + \frac{A_n}{\sqrt{1 - |A_n|^2 \lambda_n^2}} = 0. \quad (4.3)$$

Equation (4.3) can be written in the non-dimensional form

$$\ddot{A}_\star(\tau) + (\mu_0 + iv_0) \dot{A}_\star(\tau) - \Omega_1^2 A_\star(\tau) + \frac{A_\star(\tau)}{\sqrt{1 - |A_\star(\tau)|^2}} = 0, \quad (4.4)$$

where $\lambda_n A_n(\tau) = A_\star(\tau)$, $v_0 = \frac{2V_0\lambda_n}{\Omega_0}$, $\Omega_1^2 = \frac{B_0 I_0}{\lambda_n P_0} + \frac{V_0^2 \rho}{P_0}$, $\mu_0 = \frac{\eta}{\Omega_0}$, $\Omega_0^2 = \frac{P_0 \lambda_n^2}{\rho}$. If we denote $x_1(\tau) = \text{Re}A_\star(\tau)$, $x_2 = \text{Im}A_\star(\tau)$, $x_3 = \text{Re}\dot{A}_\star(\tau)$, and $x_4 = \text{Im}\dot{A}_\star(\tau)$, then the differential equation (4.4) will obtain the form

$$\begin{cases} \dot{x}_1 = x_3 \\ \dot{x}_2 = x_4 \\ \dot{x}_3 = -\mu_0 x_3 + v_0 x_4 + \Omega_1^2 x_1 - \frac{x_1}{\sqrt{1 - (x_1^2 + x_2^2)}} \\ \dot{x}_4 = -\mu_0 x_4 - v_0 x_3 + \Omega_1^2 x_2 - \frac{x_2}{\sqrt{1 - (x_1^2 + x_2^2)}} \end{cases} \quad (4.5)$$

The equilibrium points for the system (4.5) are as follows:

- 1) If $\Omega_1^2 - 1 < 0$, there is only one equilibrium point, $M_0(x_1 = 0, x_2 = 0, x_3 = 0, x_4 = 0)$.
- 2) If $\Omega_1^2 - 1 > 0$ the equilibrium points are

$$M_1(x_1 = 0, x_2 = 0, x_3 = 0, x_4 = 0), \quad M_\star^\pm(x_1 = 0, x_2 = \pm x_0, x_3 = 0, x_4 = 0), \text{ or}$$

$$M_{\star\star}^\pm(x_1 = \pm x_0, x_2 = 0, x_3 = 0, x_4 = 0), \text{ where } x_0 = \sqrt{\frac{\Omega_1^4 - 1}{\Omega_1^4}}.$$

To investigate behavior of the solution in the vicinity of the obtained points in the general case, we must solve and study the fourth order equation. Representing $A_\star(\tau)$ in polar form, $A_\star(\tau) = \rho(\tau)e^{i\theta(\tau)}$ and separating the real and imaginary parts of equation (4.4), we obtain

$$\begin{cases} \ddot{\rho} - \rho\dot{\theta}^2 + \mu_0\dot{\rho} - v_0\rho\dot{\theta} - \Omega_1^2\rho + \frac{\rho}{\sqrt{1 - \rho^2}} = 0 \\ 2\dot{\rho}\dot{\theta} + \rho\ddot{\theta} + v_0\dot{\rho} + \mu_0\rho\dot{\theta} = 0 \end{cases} \quad (4.6)$$

In the particular case when $\mu_0 = 0$, it is easy to see that if $\theta = -\frac{v_0}{2}\tau + \theta_1$ ($\theta_1 = \text{constant}$) then the second equation in (4.6) becomes an identity and the first equation has the form

$$\ddot{\rho} + \rho\left(\frac{v_0^2}{4} - \Omega_1^2\right) + \frac{\rho}{\sqrt{1 - \rho^2}} = 0. \quad (4.7)$$

Equation (4.7) has the following equilibrium points:

- 1) If $n_0 + 1 > 0$ there is only one equilibrium point $M_0(x_1 = 0, x_2 = 0)$, where $n_0 = \frac{v_0^2}{4} - \Omega_1^2$.
- 2) If $n_0 + 1 < 0$ we have $M_1(x_1 = 0, x_2 = 0)$, $M_2(x_1 = \pm\sqrt{x_0}, x_2 = 0)$, where $x_0 = \frac{n_0^2 - 1}{n_0^2}$.

The point M_0 is a center and M_1 is a saddle point. The point $M_2(x_1 = \pm\sqrt{x_0}, x_2 = 0)$ is a center (i.e. phase curves in this case have the form $\dot{\rho}^2 + n_0\rho^2 - 2\sqrt{1 - \rho^2} = \text{constant}$).

We can investigate the behavior of the string through numerical calculations of equation (4.4). Figures 4.1-4.4 consider the case $v_0 = \frac{2V_0\lambda_n}{\Omega_0} = 0$, and $\Omega_1 = \sqrt{\frac{B_0I_0}{\lambda_n P_0} + \frac{V_0^2\rho}{P_0}} = 0.1$. Figures 4.5-4.8 assume $v_0 = \frac{2V_0\lambda_n}{\Omega_0} = 0$, and $\Omega_1 = \sqrt{\frac{B_0I_0}{\lambda_n P_0} + \frac{V_0^2\rho}{P_0}} = 1.1$. Figures 4.9-4.12 consider $v_0 = \frac{2V_0\lambda_n}{\Omega_0} = 0.2$, and $\Omega_1 = \sqrt{\frac{B_0I_0}{\lambda_n P_0} + \frac{V_0^2\rho}{P_0}} = 0.1$. Figures 4.13-4.16 assume $v_0 = \frac{2V_0\lambda_n}{\Omega_0} = 0.3$, and $\Omega_1 = \sqrt{\frac{B_0I_0}{\lambda_n P_0} + \frac{V_0^2\rho}{P_0}} = 0.1$. Figures 4.17-4.20 assume $v_0 = \frac{2V_0\lambda_n}{\Omega_0} = 0.3$, and $\Omega_1 = \sqrt{\frac{B_0I_0}{\lambda_n P_0} + \frac{V_0^2\rho}{P_0}} = 1.42$. In each case we plot the amplitude of solutions and the phase portrait. The red line represents the perturbation in the z -direction, whereas the blue line represents the perturbation in the y -direction.

Notice the cases shown in figures 4.1, 4.2, 4.5, and 4.6 result in periodic trajectories for all time, since damping is neglected. When damping is applied, the string approaches a fixed orientation as $\tau \rightarrow \infty$ (figures 4.3, 4.4, 4.7, and 4.8). The cases shown in figures 4.9, 4.10, 4.13, 4.14, 4.17, and 4.18 have show chaotic or pre-chaotic type behavior as $\tau \rightarrow \infty$. We examine period doubling of solutions in these cases. Applying damping to these cases negates the chaotic behavior, causing the system to converge to a fixed orientation; these cases are shown in figures 4.11, 4.12, 4.15, 4.16, 4.19, and 4.20. The idea of chaotic motion is further discussed in Chapter 5.

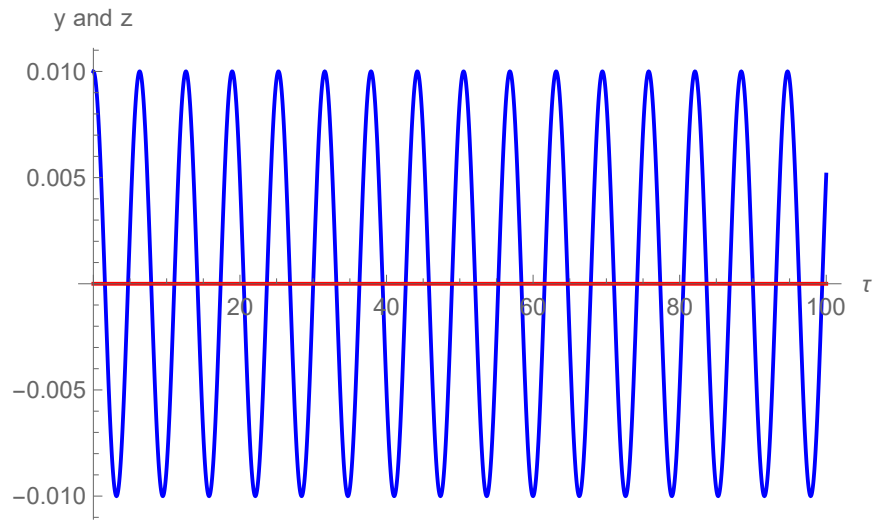


Figure 4.1: Position in y, z vs time, τ , with no damping. The red line represents the perturbation in the z -direction and the blue line represents the perturbation in the y -direction.

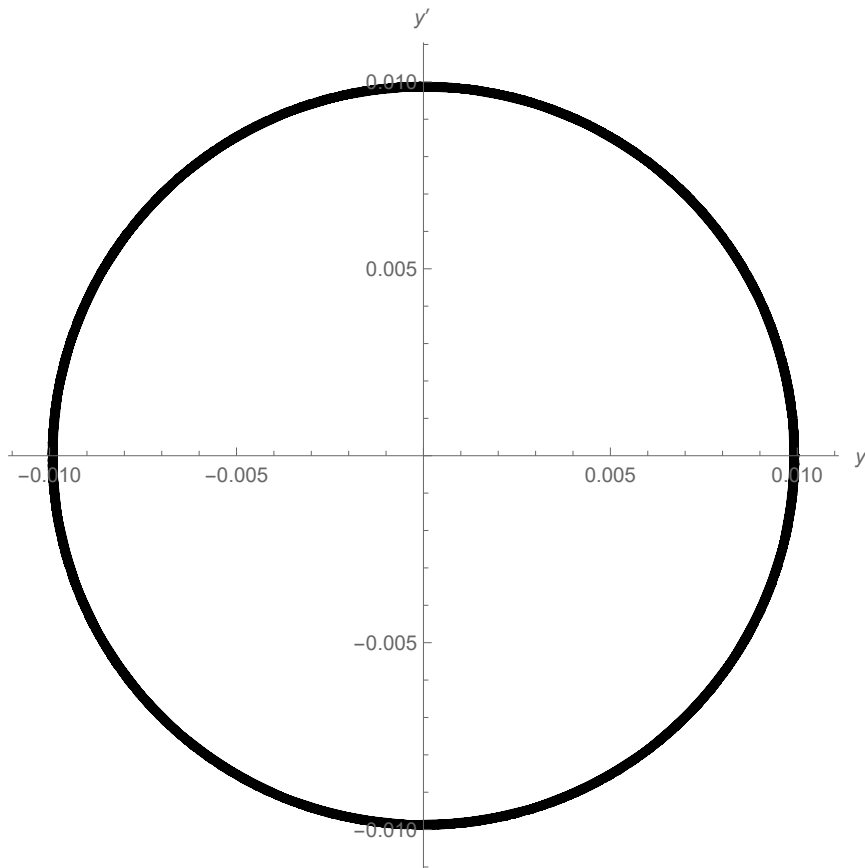


Figure 4.2: y' vs y with no damping.

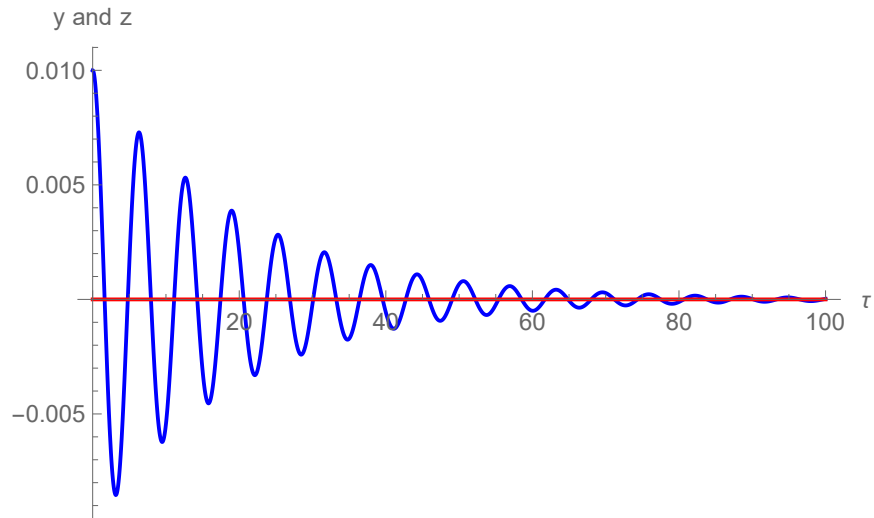


Figure 4.3: Position in y, z vs time, τ , with damping, $\mu_0 = 0.1$. The red line represents the perturbation in the z -direction and the blue line represents the perturbation in the y -direction.

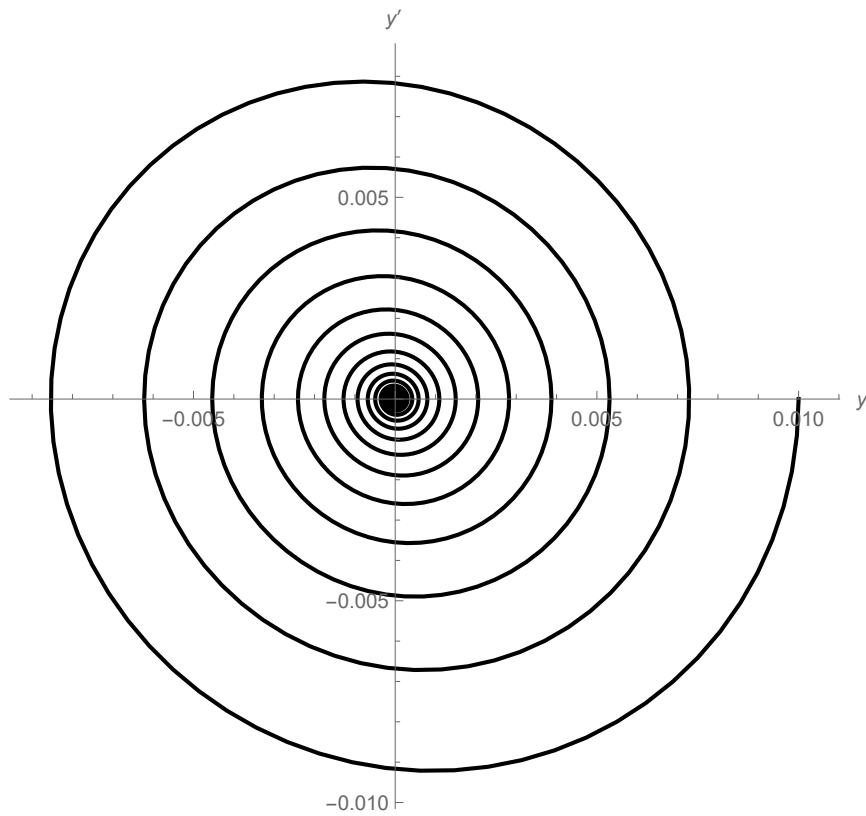


Figure 4.4: y' vs y with damping, $\mu_0 = 0.1$.

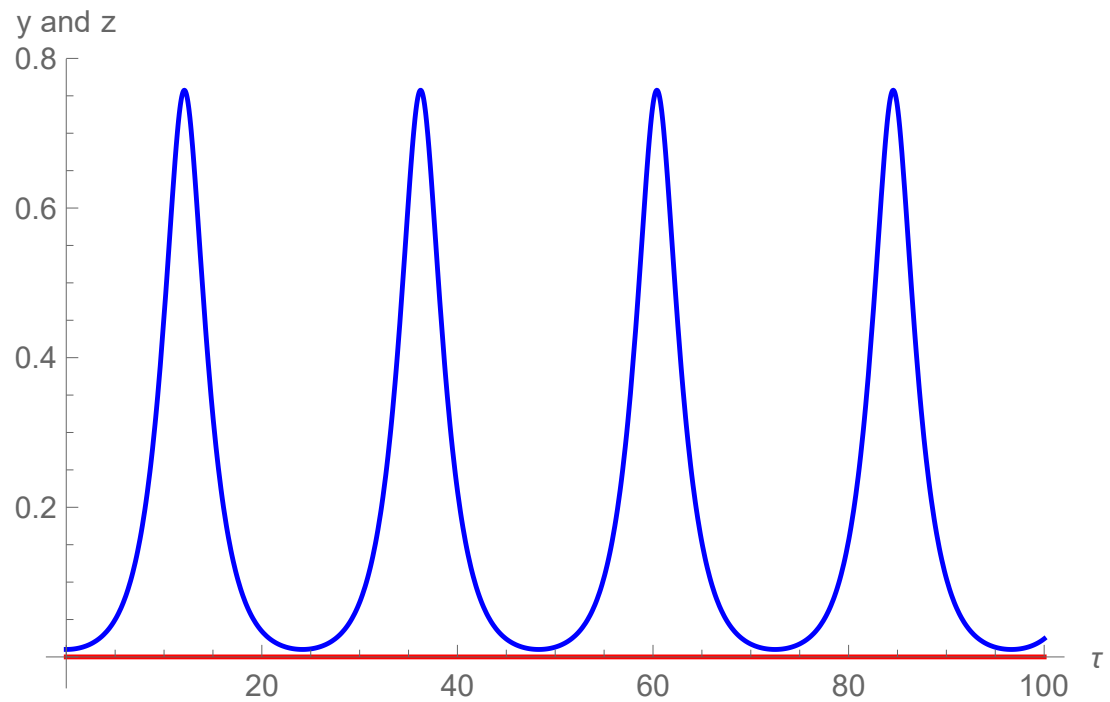


Figure 4.5: Position in y, z vs time, τ , with no damping. The red line represents the perturbation in the z -direction and the blue line represents the perturbation in the y -direction.

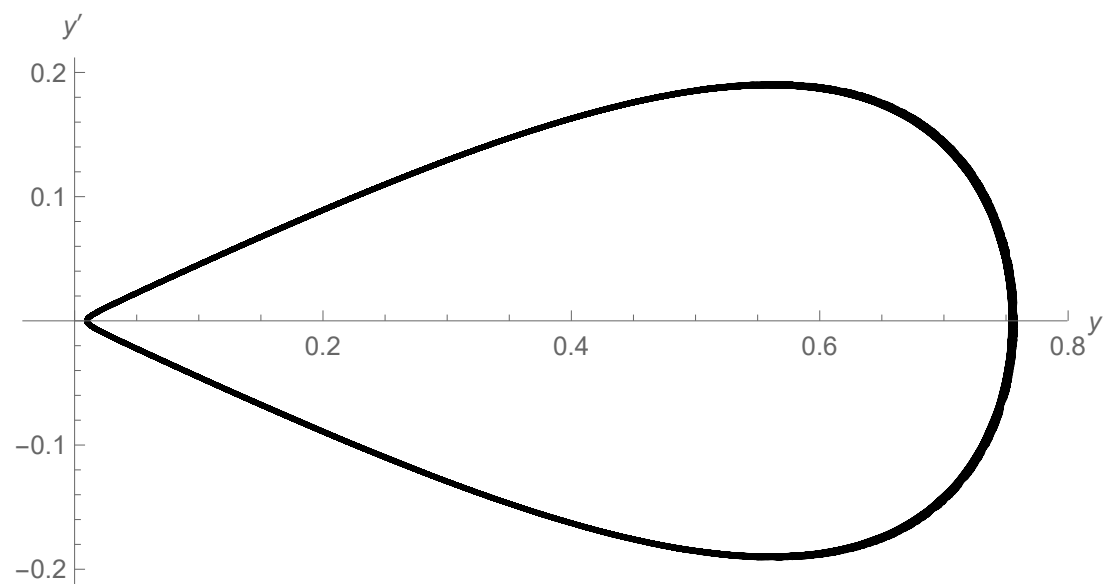


Figure 4.6: y' vs y with no damping.

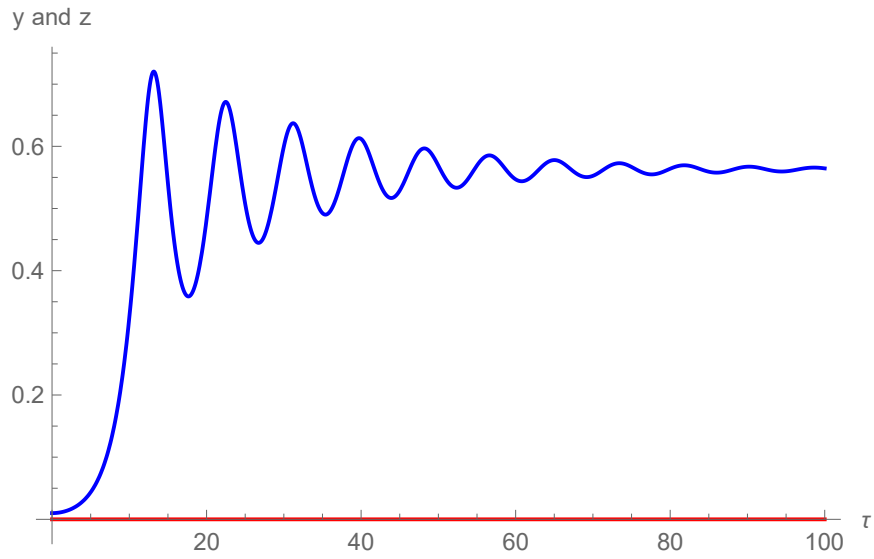


Figure 4.7: Position in y, z vs time τ , with damping $\mu_0 = 1.1$. The red line represents the perturbation in the z -direction and the blue line represents the perturbation in the y -direction.

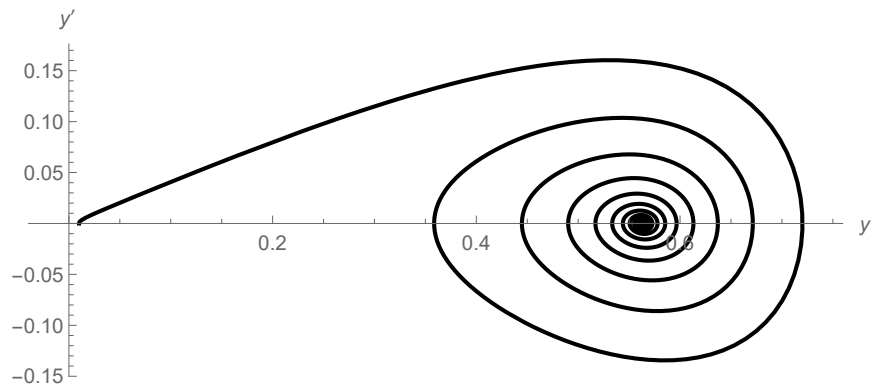


Figure 4.8: y' vs y with damping, $\mu_0 = 1.1$.

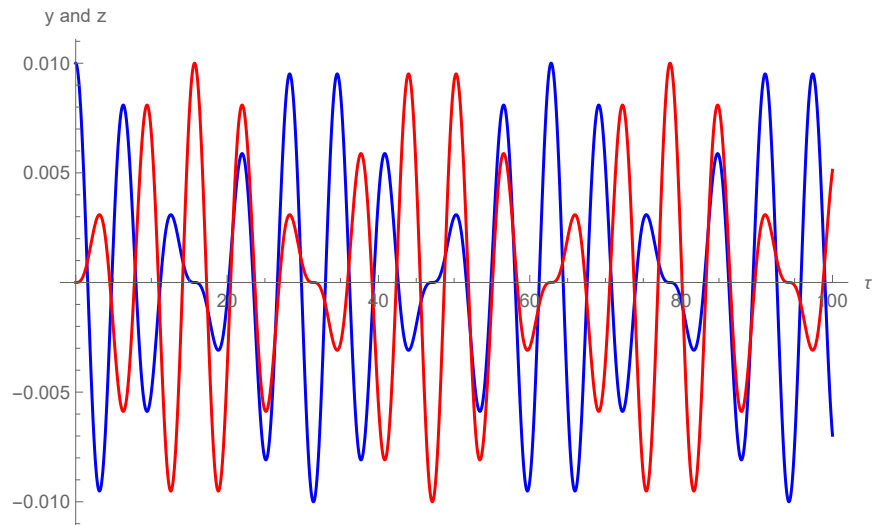


Figure 4.9: Position in y, z vs time, τ , with no damping. The red line represents the perturbation in the z -direction and the blue line represents the perturbation in the y -direction.

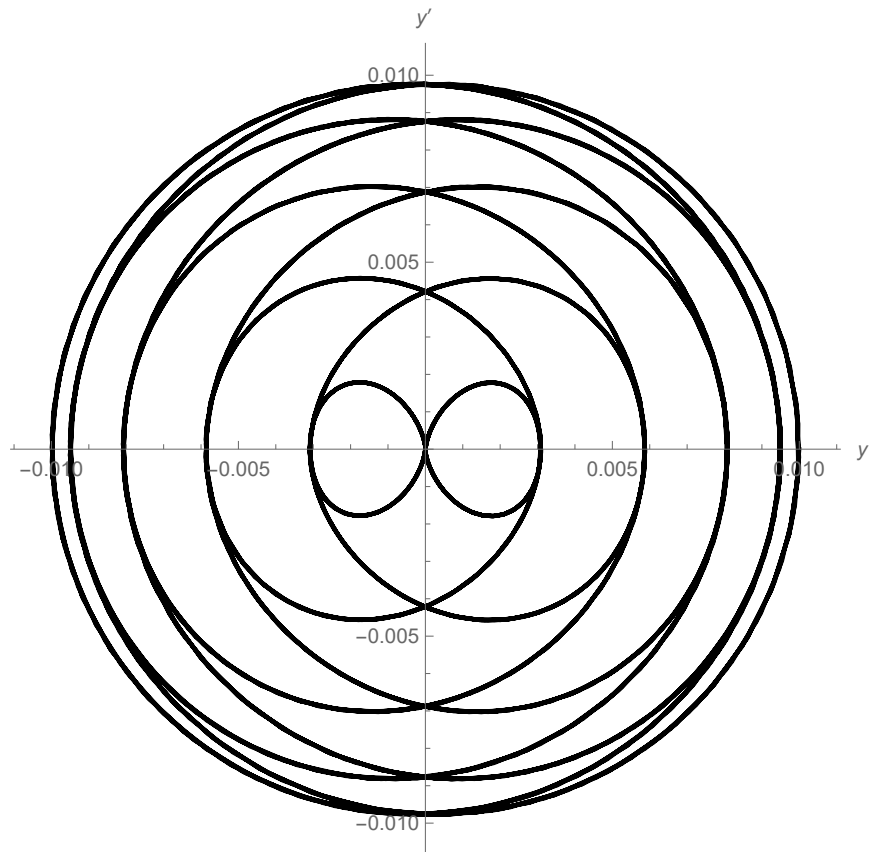


Figure 4.10: y' vs y with no damping.

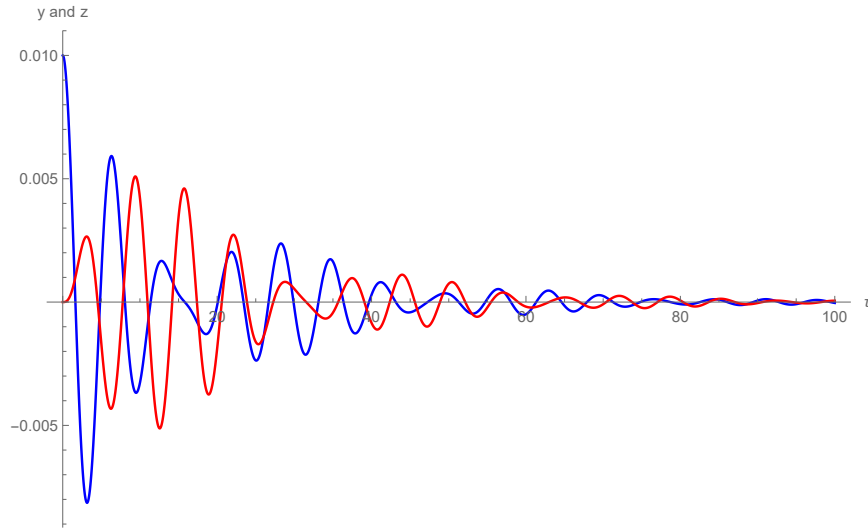


Figure 4.11: Position in y, z vs time, τ , with damping $\mu_0 = 0.1$. The red line represents the perturbation in the z -direction and the blue line represents the perturbation in the y -direction.

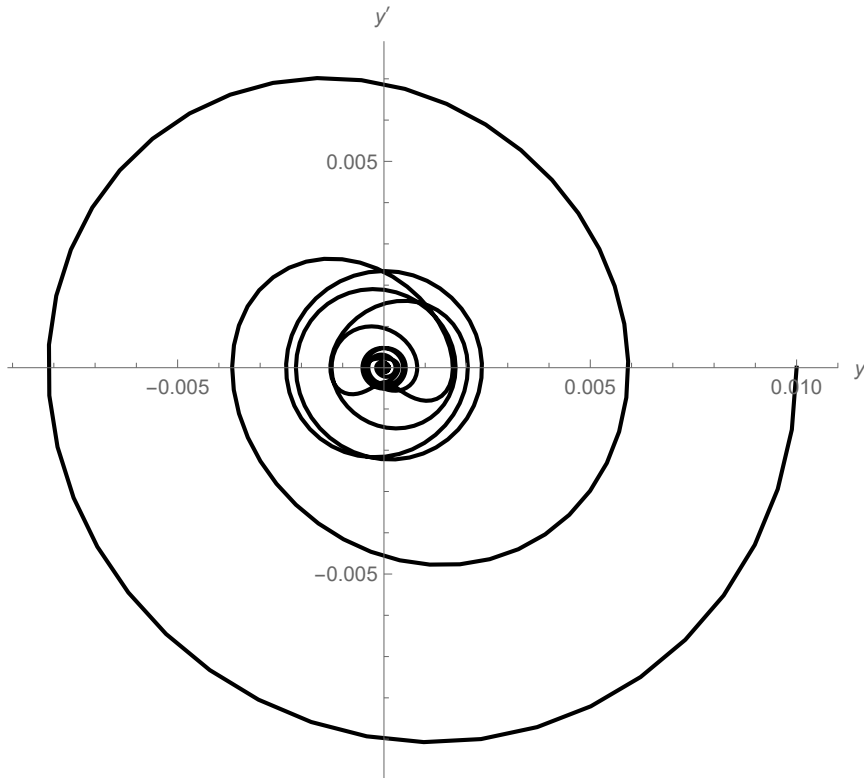


Figure 4.12: y' vs y with damping, $\mu_0 = 0.1$.

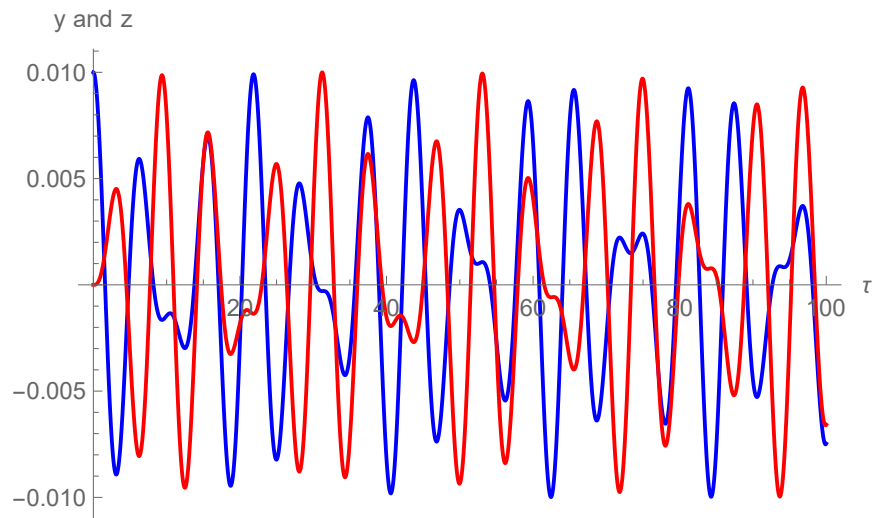


Figure 4.13: Position in y, z vs time, τ , with no damping. The red line represents the perturbation in the z -direction and the blue line represents the perturbation in the y -direction.

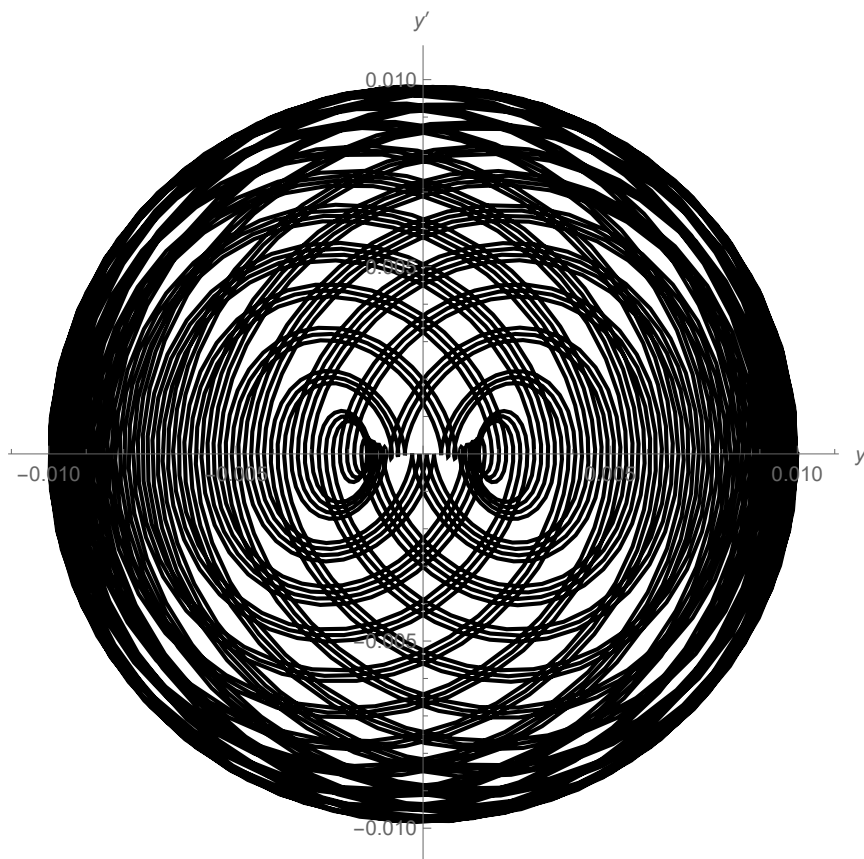


Figure 4.14: y' vs y with no damping.

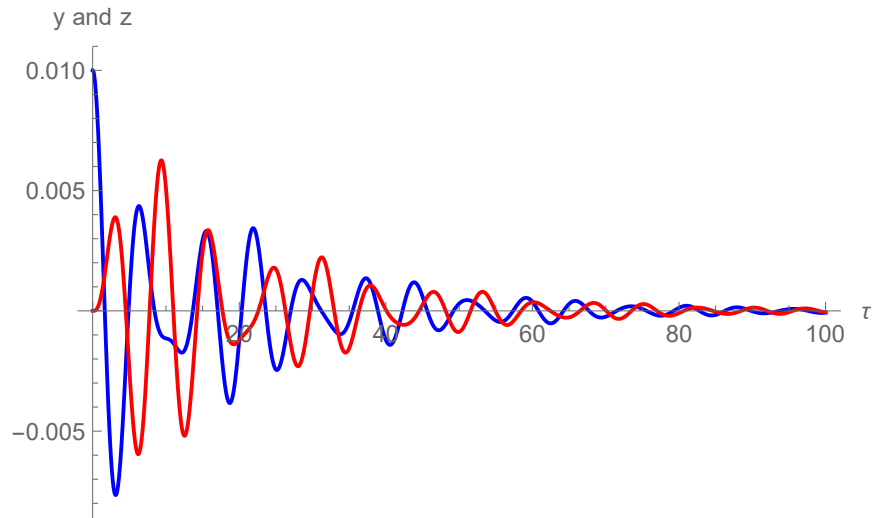


Figure 4.15: Position in y, z vs time, τ , with damping $\mu_0 = 0.1$. The red line represents the perturbation in the z -direction and the blue line represents the perturbation in the y -direction.

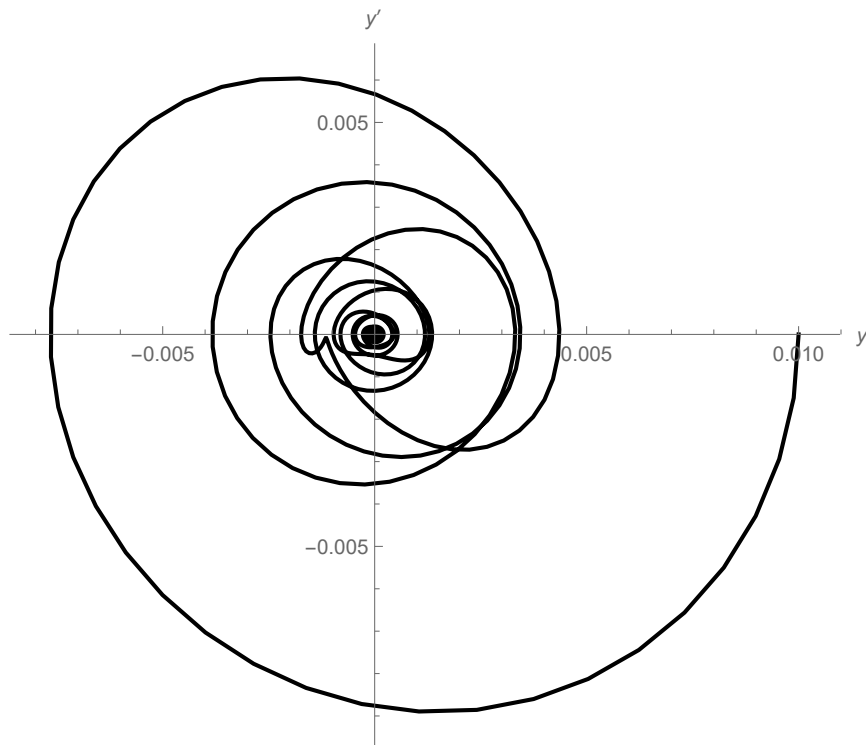


Figure 4.16: y' vs y with damping, $\mu_0 = 0.1$.

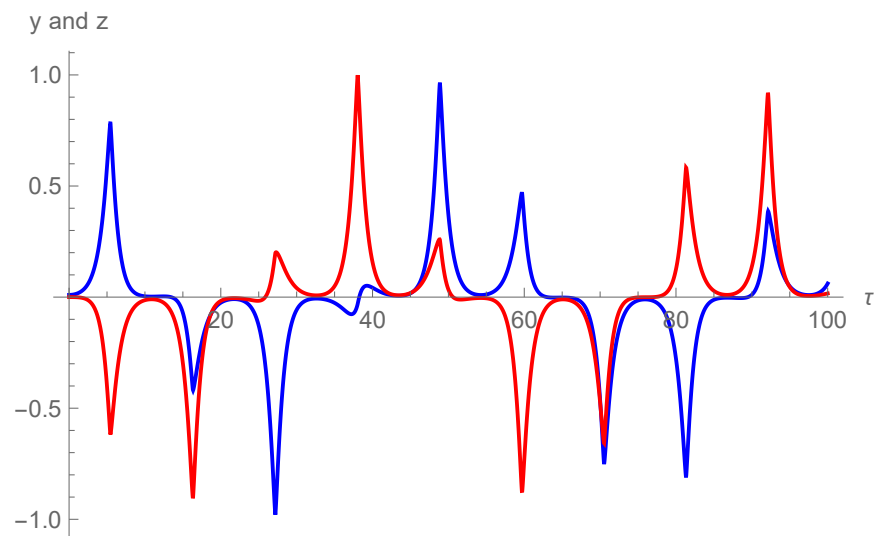


Figure 4.17: Position in y, z vs time, τ , with no damping. The red line represents the perturbation in the z -direction and the blue line represents the perturbation in the y -direction.

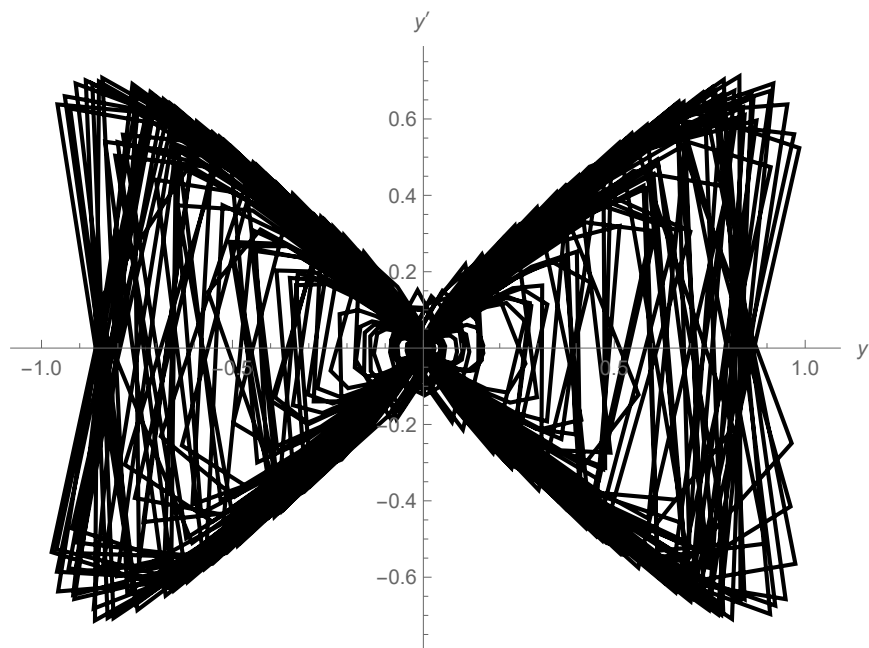


Figure 4.18: y' vs y with no damping.

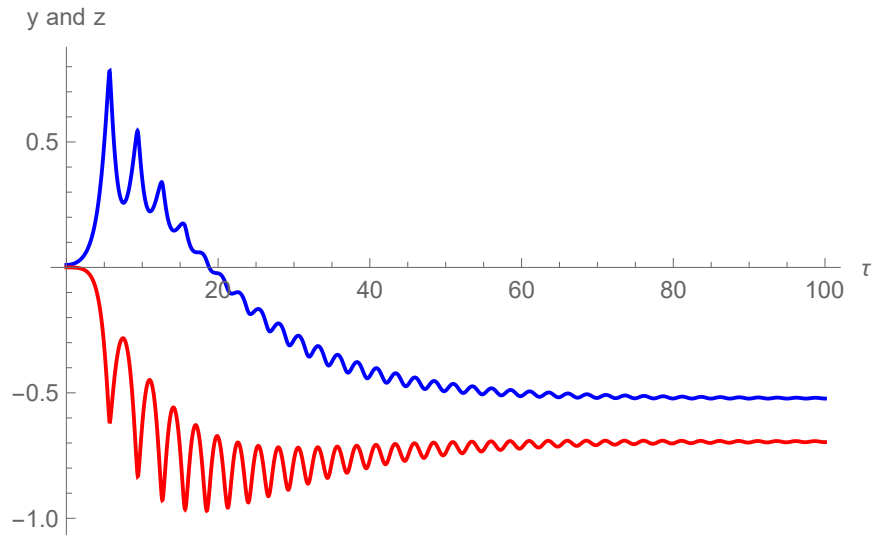


Figure 4.19: Position in y, z vs time, τ , with damping $\mu_0 = 0.1$. The red line represents the perturbation in the z -direction and the blue line represents the perturbation in the y -direction.

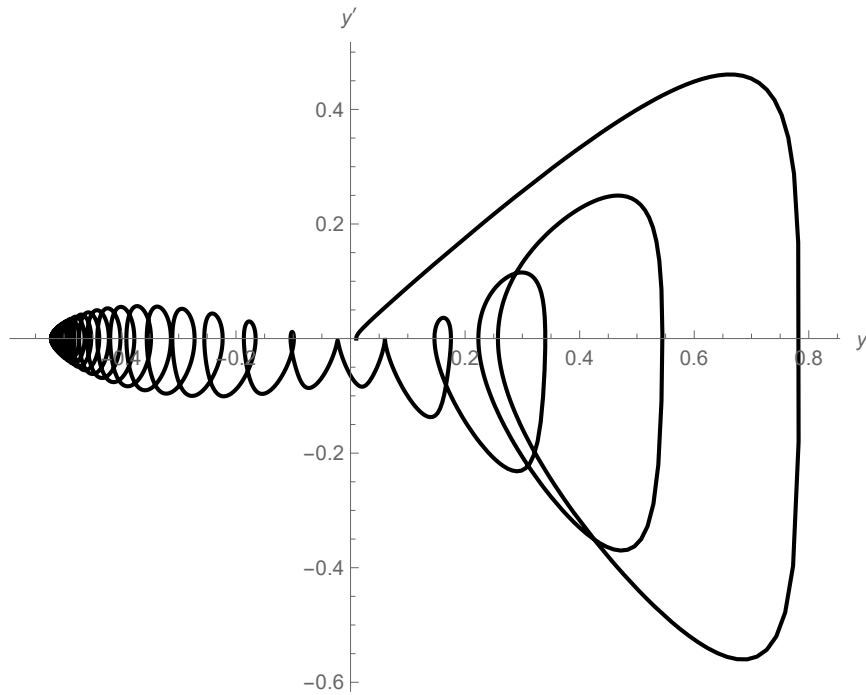


Figure 4.20: y' vs y with damping, $\mu_0 = 0.1$.

Chapter 5

Nonlinear Vibrations of a Parametrically Excited String in Resonance Cases

The goal of this chapter is to develop a criteria to characterize chaotic behavior in a string. To this end, we begin by assuming that the current flowing through the string is given by the expression

$$I_0 = I_{01} + I_{02} \cos(\Omega\tau), \quad (5.1)$$

where I_{01} , I_{02} , and Ω are constants. Replacing I_0 in (4.4) by (5.1), assuming motion with small amplitudes, $|A_\star|^2 \lambda_n^2 \ll 1$, replacing $\sqrt{1 - |A_\star|^2 \lambda_n^2} \approx 1 + |A_\star|^2 \lambda_n^2 / 2$, and introducing $X(\tau) = \frac{A_\star(\tau)}{\epsilon}$, we obtain

$$\ddot{X}(\tau) + \omega_0^2 X(\tau) = -\epsilon \left[2(\mu + iv) \dot{X}(\tau) + g \cos(\Omega\tau) X(\tau) + \alpha X(\tau) |X(\tau)|^2 \right], \quad (5.2)$$

where $\mu = \frac{\mu_0}{2\epsilon}$, $v = \frac{v_0}{\epsilon}$, $\omega_0^2 = 1 - \frac{I_{01}B_0}{\lambda_n P_0} - \frac{V_0^2 \rho}{P_0}$, $g = \frac{B_0 I_{02}}{\epsilon P_0 \lambda_m}$, $\alpha = \frac{\lambda_m^2 \epsilon}{2}$, $\epsilon \ll 1$.

In obtaining (5.2) we were limited by the cubic approximation for the amplitude of vibration. We will investigate (5.2) near resonance vibrations by considering the cases

I) $\Omega \simeq \omega_0$, and

II) $\Omega \simeq 2\omega_0$.

If we assume $\Omega \simeq \omega_0$, then we are investigating the first natural frequency of the linear system. Similarly, $\Omega \simeq 2\omega_0$ is the second natural frequency.

A system under resonance vibration will have increasing amplitudes over time. In general, there

is no criteria to categorize chaos in an arbitrary system. However, as mentioned by Moon in [12], period doubling is a route to chaotic behavior. Thus we develop criteria to identify period doubling and criteria for which the solutions near $\Omega \simeq \omega_0$ and $\Omega \simeq 2\omega_0$ are stable.

5.1 Resonance Vibrations of the String Near the Base State

In the base state $\Omega \simeq \omega_0$, and we assume

$$\Omega^2 = \omega_0^2 + \epsilon\sigma_1, \quad (5.3)$$

where σ_1 is a detuning parameter. Equation (5.2) then takes the form

$$\ddot{X} + \Omega^2 X = \epsilon[\sigma_1 X - 2(\mu + iv)\dot{X} - g \cos(\Omega\tau)X - \alpha X|X|^2]. \quad (5.4)$$

Using the Method of Multiple Scales (MMS) from [14] to approximate the solution of equation (5.4). We represent the solution of (5.4) as an expansion

$$X(\tau, \epsilon) = X_0(T_0, T_1, T_2) + \epsilon X_1(T_0, T_1, T_2) + \dots, \quad (5.5)$$

where $T_n = \epsilon^n \tau$, $n = 0, 1, \dots$. Derivatives with respect to time take the forms

$$\frac{d}{d\tau} = D_0 + \epsilon D_1 + \epsilon^2 D_2 + \dots, \quad \frac{d^2}{d\tau^2} = D_0^2 + 2\epsilon D_0 D_1 + \epsilon^2 (2\epsilon D_0 D_2 + D_1^2) + \dots, \quad (5.6)$$

where $D_n = \partial/\partial T_n$. Substituting (5.5), (5.6) into (5.4) and equating the coefficients of like ϵ , we obtain

$$D_0^2 x_0 + \Omega^2 x_0 = 0 \quad (5.7)$$

$$D_0^2 x_1 + \Omega^2 x_1 = 2\sigma x_0 D_0 D_1 x_0 - 2(\mu + iv)D_0 x_0 - \alpha \bar{x}_0 x_0^2 - \frac{g x_0}{2}(e^{-i\Omega T_0} + e^{i\Omega T_0}) \quad (5.8)$$

$$\begin{aligned}
D_0^2 x_2 + \Omega^2 x_2 = & \sigma x_1 - 2D_0 D_2 x_0 - D_1^2 x_0 - 2D_0 D_1 x_1 - 2(\mu + iv)D_0 x_1 \\
& - 2(\mu + iv)D_1 x_0 - \alpha(x_0^2 \bar{x}_1 + x_0 \bar{x}_0 x_1) - \frac{g x_1}{2}(e^{-i\Omega T_0} + e^{i\Omega T_0})
\end{aligned} \tag{5.9}$$

Consider solutions of equation (5.7) of the form

$$x_0 = A(T_1, T_2)e^{i\Omega T_0} + B(T_1, T_2)e^{-i\Omega T_0}, \tag{5.10}$$

where the coefficients A and B are functions of T_1 and T_2 at this level of approximation. These functions are determined by imposing the solvability conditions at the next level of approximation.

By substituting x_0 into equation (5.8) we obtain

$$\begin{aligned}
\frac{\partial^2 x_1}{\partial T_0^2} + \Omega^2 x_1 = & e^{i\Omega T_0} \left(\sigma_1 A - 2i\Omega \frac{\partial A}{\partial T_1} - 2(\mu + iv)i\Omega A - \alpha A|A|^2 - 2\alpha A|B|^2 \right) \\
& + e^{i\Omega T_0} \left(\sigma_1 B - 2i\Omega \frac{\partial B}{\partial T_1} - 2(\mu + iv)i\Omega B - \alpha B|B|^2 - 2\alpha B|A|^2 \right) \\
& - \frac{g(A + B)}{2} - \alpha A^2 \bar{B} e^{3i\Omega T_0} - \alpha B^2 \bar{A} e^{-3i\Omega T_0} - \frac{gA}{2} e^{2i\Omega T_0} - \frac{gB}{2} e^{-2i\Omega T_0}.
\end{aligned} \tag{5.11}$$

Depending on the functions A and B the particular solutions of equation (5.11) may contain secular terms. The conditions to eliminate these secular terms (the solvability condition) is

$$\sigma_1 A - 2i\Omega \frac{\partial A}{\partial T_1} - 2(\mu + iv)i\Omega A - \alpha A|A|^2 - 2\alpha A|B|^2 = 0, \tag{5.12}$$

$$\sigma_1 B - 2i\Omega \frac{\partial B}{\partial T_1} - 2(\mu + iv)i\Omega B - \alpha B|B|^2 - 2\alpha B|A|^2 = 0. \tag{5.13}$$

A particular solution of equation (5.11) is

$$x_1 = \frac{g(A + B)}{2\Omega^2} + \frac{\alpha A^2 \bar{B}}{2\Omega^2} e^{3i\Omega \tau} + \frac{\alpha B^2 \bar{A}}{8\Omega^2} e^{-3i\Omega \tau} + \frac{gA}{6\Omega^2} e^{2i\Omega \tau} + \frac{gB}{6\Omega^2} e^{-2i\Omega \tau}. \tag{5.14}$$

Substituting equations (5.10) and (5.14) into (5.9) and eliminating secular terms from x_2 , results in

$$\begin{aligned} & -2\Omega \frac{\partial A}{\partial T_2} + \left[(\mu + iv)^2 + \frac{\sigma_1^2}{4\Omega^2} + \frac{g^2}{6\Omega^2} \right] A + \frac{g^2 B}{4\Omega^2} \\ & + \frac{\alpha}{2i\Omega} \left[-2\mu + iv - \frac{i\sigma_1}{\Omega} \right] A [|A|^2 + 2|B|^2] + \frac{\alpha^2 A}{8\Omega^2} [2|A|^4 + 6|AB|^2 + 7|B|^4] = 0, \end{aligned} \quad (5.15)$$

$$\begin{aligned} & -2i\Omega \frac{\partial B}{\partial T_2} + \left[(\mu + iv)^2 + \frac{\sigma_1^2}{4\Omega^2} + \frac{g^2}{6\Omega^2} \right] B + \frac{g^2 A}{4\Omega^2} \\ & + \frac{\alpha}{2i\Omega} \left[2\mu - \frac{i\sigma_1}{\Omega} \right] B [|B|^2 + 2|A|^2] + \frac{\alpha^2 B}{8\Omega^2} [2|B|^4 + 6|AB|^2 + 7|A|^4] = 0. \end{aligned} \quad (5.16)$$

To solve equations (5.12), (5.13), and (5.16), we combine them into one first order ordinary differential equation. This procedure is usually referred to as the method of reconstitution and involves finding the ordinary differential equation that yields equations (5.12)–(5.16) when analyzed by the MMS. The result is

$$\begin{aligned} & -2i\Omega \dot{A} + \epsilon \left\{ \sigma_1 - 2i\Omega(\mu + iv) - \alpha [|A|^2 + 2|B|^2] \right\} A + \epsilon^2 \left\{ \left[(\mu + iv)^2 + \frac{\sigma_1^2}{4\Omega^2} + \frac{g^2}{6\Omega^2} \right] A \right. \\ & \left. + \frac{g^2 B}{4\Omega^2} + \frac{\alpha}{2i\Omega} \left[-2\mu + iv - \frac{i\sigma_1}{\Omega} \right] A [|A|^2 + 2|B|^2] + \frac{\alpha^2 A}{8\Omega^2} [2|A|^4 + 6|AB|^2 + 7|B|^4] \right\} = 0, \end{aligned} \quad (5.17)$$

$$\begin{aligned} & 2i\Omega \dot{B} + \epsilon \left\{ \sigma_1 + 2i\Omega(\mu + iv) - \alpha [|B|^2 + 2|A|^2] \right\} B + \epsilon^2 \left\{ \left[(\mu + iv)^2 + \frac{\sigma_1^2}{4\Omega^2} + \frac{g^2}{6\Omega^2} \right] B + \frac{g^2 A}{4\Omega^2} \right. \\ & \left. + \frac{\alpha}{2i\Omega} \left[2\mu - \frac{i\sigma_1}{\Omega} \right] B [|B|^2 + 2|A|^2] + \frac{\alpha^2 B}{8\Omega^2} [2|B|^4 + 6|AB|^2 + 7|A|^4] \right\} = 0. \end{aligned} \quad (5.18)$$

Substituting Equation (5.10) and (5.14) into (5.5) we find a second order approximation to the solution of (5.4)

$$\begin{aligned} X(\tau, \epsilon) \equiv X_{\omega_0}(\tau, \epsilon) &= Ae^{i\Omega\tau} + Be^{-i\Omega\tau} \\ &+ \epsilon \left\{ -\frac{g(A+B)}{2\Omega^2} + \frac{\alpha A^2 \bar{B}}{8\Omega^2} e^{3i\Omega\tau} + \frac{\alpha B^2 \bar{A}}{8\Omega^2} e^{-3i\Omega\tau} + \frac{gA}{6\Omega^2} e^{2i\Omega\tau} + \frac{gB}{6\Omega^2} e^{-2i\Omega\tau} \right\} + O(\epsilon^2). \end{aligned} \quad (5.19)$$

The stationary regime, which represents a limit circle or auto-oscillation regime, is obtained from (5.17) and (5.18) when $\dot{A} = \dot{B} = 0$. Note that the equation (5.4), describing the stationary regime, can be solved numerically.

5.2 Resonance Vibrations of the String Near the Main State

Now consider $\Omega \simeq 2\omega_0$. We introduce the detuning parameter σ_2 :

$$\Omega^2 = 4(\omega_0^2 + \epsilon\sigma_2), \quad (5.20)$$

Similar to the solution $X(t)$ in the base case, we obtain

$$X(\tau, \epsilon) \equiv X_{2\omega_0}(\tau, \epsilon) = Ae^{\frac{i\Omega\tau}{2}} + Be^{-\frac{i\Omega\tau}{2}} + \epsilon \left\{ \left(\frac{\alpha A^2 \bar{B}}{2\Omega^2} + \frac{gA}{4\Omega^2} \right) e^{\frac{3i\Omega\tau}{2}} + \left(\frac{\alpha B^2 \bar{A}}{2\Omega^2} + \frac{gB}{4\Omega^2} \right) e^{-\frac{3i\Omega\tau}{2}} \right\} + O(\epsilon^2). \quad (5.21)$$

Functions A and B are defined in the first approximation by

$$-i\Omega\dot{A} + \epsilon \left\{ [\sigma_2 - i\Omega(\mu + iv) - \alpha(|A|^2 + 2|B|^2)]A - \frac{gB}{2} \right\} + O(\epsilon^2) = 0 \quad (5.22)$$

$$i\Omega\dot{B} + \epsilon \left\{ [\sigma_2 + i\Omega(\mu + iv) - \alpha(|B|^2 + 2|A|^2)]B - \frac{gB}{2} \right\} + O(\epsilon^2) = 0. \quad (5.23)$$

The stationary or stable regime is obtained from equations (5.22) and (5.23) when $\dot{A} = \dot{B} = 0$.

5.3 Stability of Solutions Near Base State and Main State

Recall the base and main states are considering $\Omega \simeq \omega_0$ and $\Omega \simeq 2\omega_0$, respectively. In sections 5.1 and 5.2, solutions $X_{\omega_0}(\tau, z)$ and $X_{2\omega_0}(\tau, z)$ were obtained from equation (5.2) when Ω is close to ω_0 and $2\omega_0$, respectively. Both solutions depend on A and B , which in turn satisfy equations (5.17) and (5.18), or (5.22) and (5.23). Steady state periodic solutions correspond to $\dot{A} = \dot{B} = 0$, the stationary points of these equations. Note that the trivial solutions with $A = B = 0$ is a possible stationary point (straight-line form of the string) for equations (5.17), (5.18) and (5.22), (5.23).

Let's now investigate the stability of the trivial solution for equations (5.17) and (5.18). After neglecting nonlinear terms, the equations take the form

$$-2i\Omega\dot{A} + \epsilon[\sigma_1 - 2i\Omega(\mu + iv)]A + \epsilon^2 \left[(\mu + iv)^2 + \frac{\sigma_1^2}{4\Omega^2} + \frac{g^2}{4\Omega^2} \right] A + \frac{\epsilon^2 g^2 B}{4\Omega^2} = 0, \quad (5.24)$$

$$-2i\Omega\dot{B} + \epsilon[\sigma_1 - 2i\Omega(\mu + iv)]B + \epsilon^2\left[(\mu + iv)^2 + \frac{\sigma_1^2}{4\Omega^2} + \frac{g^2}{4\Omega^2}\right]B + \frac{\epsilon^2 g^2 A}{4\Omega^2} = 0. \quad (5.25)$$

Separating the real and imaginary parts in equations (5.1) and (5.2) where, $A = A_r + iA_i$, $B = B_r + iB_i$, results in four linear differential equations with respect to A_r, A_i, B_r, B_i . Considering solutions of the form

$$(A_r, A_i, B_r, B_i) = (a_r, a_i, b_r, b_i)e^{\epsilon\lambda t}, \quad (5.26)$$

results in an algebraic equation for the parameter λ :

$$\Delta_{\omega_0}(\lambda, g, \Omega, \mu, \gamma) = \det \hat{Q} = 0, \quad (5.27)$$

where $Q_{11} = -Q_{22} = 2\Omega\lambda + l_1$, $Q_{13} = Q_{24} = Q_{31} = Q_{42} = 0$, $Q_{14} = Q_{23} = -Q_{32} = -Q_{41} = -r_1$,
 $Q_{21} = Q_{12} = -r_0 - l_0$, $Q_{43} = Q_{34} = r_0 - l_0$, $Q_{33} = -Q_{44} = 2\Omega\lambda + l_2$, $r_1 = \frac{\epsilon g^2}{4\Omega^2}$,
 $r_0 = \sigma_1 + \frac{\epsilon\sigma_1^2}{4\Omega^2} + \epsilon\mu^2 + \frac{\epsilon g^2}{6\Omega^2} - \epsilon v^2$, $l_0 = 2v\Omega$, $l_1 = 2\mu(\Omega - \epsilon v)$, and $l_2 = 2\mu(\Omega + \epsilon v)$.

The trivial solution is stable when $\text{Re}(\lambda) < 0$. Instability starts when $\text{Re}(\lambda) > 0$. The line $\text{Re}(\lambda) = 0$ describes a nontrivial curve in the plane (g, Ω) which separates the stable and unstable domains.

When $v = 0$ the nontrivial curve has the form

$$l_{\omega_0} = \left(\frac{\epsilon g^2}{4\Omega^2}\right)^2 - \left(\sigma_1 + \frac{\epsilon\sigma_1^2}{4\Omega^2} + \epsilon\mu^2 + \frac{\epsilon g^2}{6\Omega^2}\right)^2 - 4\Omega^2\mu^2 = 0. \quad (5.28)$$

An analogous investigation of the stability of the trivial solution near frequency $\Omega = 2\omega_0$ results in the characteristic equation

$$\Delta_{2\Omega_0}(\lambda, g, \Omega, \mu, \gamma) = \det \hat{R} = 0, \quad (5.29)$$

where $R_{11} = -R_{22} = -\Omega\lambda - \gamma_2$, $R_{12} = R_{21} = \gamma_1$, $R_{14} = R_{23} = R_{32} = R_{41} = -\frac{g}{2}$,
 $R_{13} = R_{31} = R_{24} = R_{42} = 0$, $R_{33} = -R_{44} = \Omega\lambda + R_1 + 2\mu\Omega$, $R_{34} = R_{43} = R_0 - 2v\Omega$,
 $R_0 = \sigma_2 + v\Omega + \epsilon\left(\mu^2 - v^2 + \frac{\sigma_2^2}{\Omega^2} - \frac{3g^2}{8r^2}\right)$, and $R_1 = -\mu(2\epsilon v + \Omega)$. The nontrivial curve is described by

$$I_{2\omega_0} = [\Omega^2\mu^2 + \delta_2^2]^2 - \left(\frac{g}{2}\right)^2 \left\{ 2(\Omega^2\mu^2 + \delta_2^2) - \left(\frac{g}{2}\right)^2 \right\} = 0, \quad (5.30)$$

where $\delta_1 = \sigma_1 + \epsilon \left(\mu^2 + \frac{\sigma_1^2}{4\Omega^2} + \frac{g^2}{6\Omega^2} \right)$, $\delta_2 = \sigma_2 + \epsilon \left(\mu^2 + \frac{\sigma_2^2}{4\Omega^2} - \frac{g^2}{6\Omega^2} \right)$.

The approximate curves in the (g, Ω) plane are shown in figure 5.1. The instability domain is shown by dashed lines. The curves $l_{\omega_0} = 0$ and $l_{2\omega_0} = 0$ are bifurcation curves. The curves in figure 5.1 are for the case where $\sigma_1 = \sigma_2 = \mu = 0.1$ and $\epsilon = 0.01$.

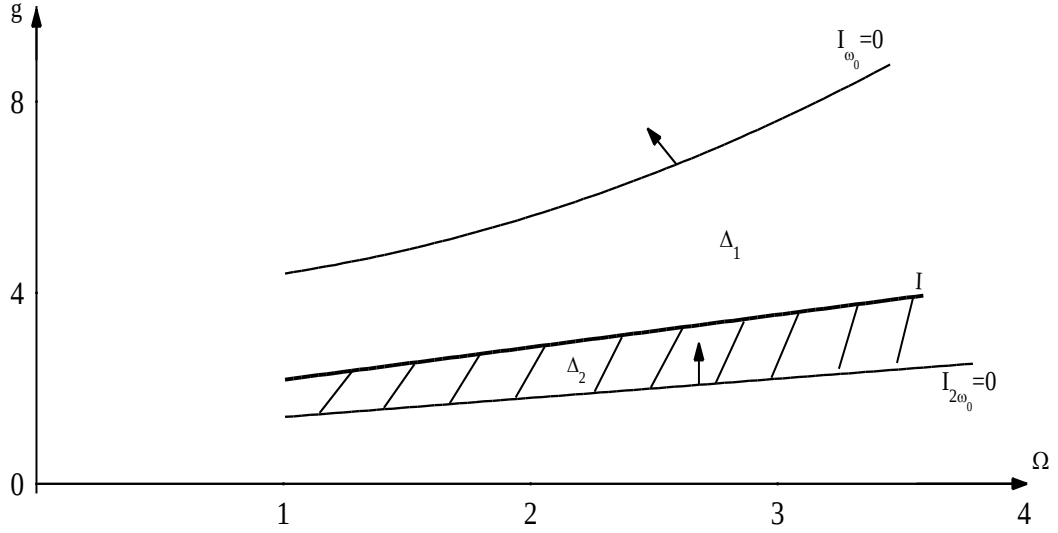


Figure 5.1: Instability regions for Δ_1 near $\Omega \simeq \omega_0$ and Δ_2 near $\Omega \simeq 2\omega_0$. This plot assumes $\sigma_1 = \sigma_2 = \mu = 0.1$ and $\epsilon = 0.01$.

The domain Δ_1 is an unstable domain near the frequency $\Omega = \omega_0$, whereas Δ_2 is an unstable domain near the frequency $\Omega = 2\omega_0$. As previously mentioned, when the frequency of periodic vibration doubles, a pre-chaotic state is expected, Moon [12]. Thus we conclude that the domain Δ_2 is a region for the parameters leading a pre-chaotic state.

The pre-chaotic behavior in the sense of frequency vibrations has been analyzed. Domains of parameters for chaotic behavior have been defined.

Chapter 6

Traveling Wave Solutions

When an infinite string is perturbed (or plucked) from an initial straight-line form, a wave is propagated through the string by subsequent particle interactions. A crest and trough travel through the string in both axial directions from the point of perturbation. The wave or pattern that travels through the infinite string is called a traveling wave. In this chapter, we assume traveling wave solutions for our system in an infinite setting. We will use the traveling wave solutions to compute corresponding wave speeds. After linearizing equation (2.9), we consider two cases; the first neglects viscosity and axial motion, where as the second only neglects viscosity.

6.1 Dead Interval for wavelength

To compute wave speeds, we will treat the nonlinearity in equation (2.9) and find the traveling wave solutions. First consider the full nonlinear equation (2.9)

$$\rho \left[\frac{\partial^2 u(s, t)}{\partial t^2} + \eta \frac{\partial u(s, t)}{\partial t} + 2V_0 \frac{\partial^2 u(s, t)}{\partial s \partial t} + V_0^2 \frac{\partial^2 u(s, t)}{\partial s^2} \right] = \frac{\partial}{\partial s} \left(\frac{P_0 \frac{\partial u(s, t)}{\partial s}}{\sqrt{1 - \left| \frac{\partial u(s, t)}{\partial s} \right|^2}} \right) - iI_0 B_0 \frac{\partial u(s, t)}{\partial s}.$$

The nonlinearity in equation (2.9) is evaluated as follows:

$$\frac{\partial}{\partial s} \left(\frac{P_0 \frac{\partial u}{\partial s}}{\sqrt{1 - \left| \frac{\partial u}{\partial s} \right|^2}} \right) = P_0 \frac{\partial^2 u}{\partial s^2} \frac{1}{\sqrt{1 - \left| \frac{\partial u}{\partial s} \right|^2}} - \frac{P_0}{2} \frac{\partial u}{\partial s} \left[1 - \left| \frac{\partial u}{\partial s} \right|^2 \right]^{-3/2} \left[\frac{\partial \bar{u}}{\partial s} \frac{\partial^2 u}{\partial s^2} + \frac{\partial u}{\partial s} \frac{\partial^2 \bar{u}}{\partial s^2} \right] \quad (6.1)$$

where $\bar{u}$ is the conjugate of u and $|\frac{\partial u}{\partial s}|^2 = \frac{\partial u}{\partial s} \frac{\partial \bar{u}}{\partial s}$. Applying Taylor series approximation to equation (6.1) leads to

$$\begin{aligned} \frac{\partial}{\partial s} \left(\frac{P_0 \frac{\partial u}{\partial s}}{\sqrt{1 - |\frac{\partial u}{\partial s}|^2}} \right) &\simeq P_0 \frac{\partial^2 u}{\partial s^2} \left[1 + \frac{1}{2} \frac{\partial u}{\partial s} \frac{\partial \bar{u}}{\partial s} \right] + \frac{P_0}{2} \frac{\partial u}{\partial s} \left[\frac{\partial \bar{u}}{\partial s} \frac{\partial^2 u}{\partial s^2} + \frac{\partial u}{\partial s} \frac{\partial^2 \bar{u}}{\partial s^2} \right] \left[1 + \frac{3}{2} \frac{\partial u}{\partial s} \frac{\partial \bar{u}}{\partial s} \right] \\ &= P_0 \frac{\partial^2 u}{\partial s^2} + P_0 \frac{\partial u}{\partial s} \frac{\partial \bar{u}}{\partial s} \frac{\partial^2 u}{\partial s^2} + \frac{P_0}{2} \left(\frac{\partial u}{\partial s} \right)^2 \frac{\partial^2 \bar{u}}{\partial s^2} + \mathcal{O}(u^5). \end{aligned} \quad (6.2)$$

Using equation (6.2) in (2.9) and neglecting high order terms gives

$$\frac{\partial^2 u}{\partial t^2} - c_0^2 \frac{\partial^2 u}{\partial s^2} + i\lambda \frac{\partial u}{\partial s} + \eta \frac{\partial u}{\partial t} + 2V_0 \frac{\partial^2 u}{\partial s \partial t} - P_0 \frac{\partial u}{\partial s} \frac{\partial \bar{u}}{\partial s} \frac{\partial^2 u}{\partial s^2} - \frac{P_0}{2} \left(\frac{\partial u}{\partial s} \right)^2 \frac{\partial^2 \bar{u}}{\partial s^2} = 0 \quad (6.3)$$

where $c_0^2 = \frac{P_0}{\rho} - V_0^2$, and $\lambda = \frac{I_0 B_0}{\rho}$.

To compute the wave speed corresponding to traveling wave solutions. We consider the case with no axial motion, $V_0 = 0$, and no viscosity, $\eta = 0$. So we are considering a current carrying string in a magnetic field (the case considered in [11]). Traveling wave solutions have the form $u(s, t) \sim u_0 e^{i(\omega t + ks)}$. Neglecting η and V_0 , the wave speed is given by

$$v = \sqrt{c_0^2 - \frac{\lambda}{k}}. \quad (6.4)$$

This implies there is a cut-off region where waves do not propagate. The wave speed relationship is shown in figure 6.1. Waves can only be propagated when $k > \frac{\lambda}{c_0^2}$. If the wavelength, $k \in [0, \frac{\lambda}{c_0^2}]$ there are no waves produced. Physically, the magnetic field is preventing wave formation.

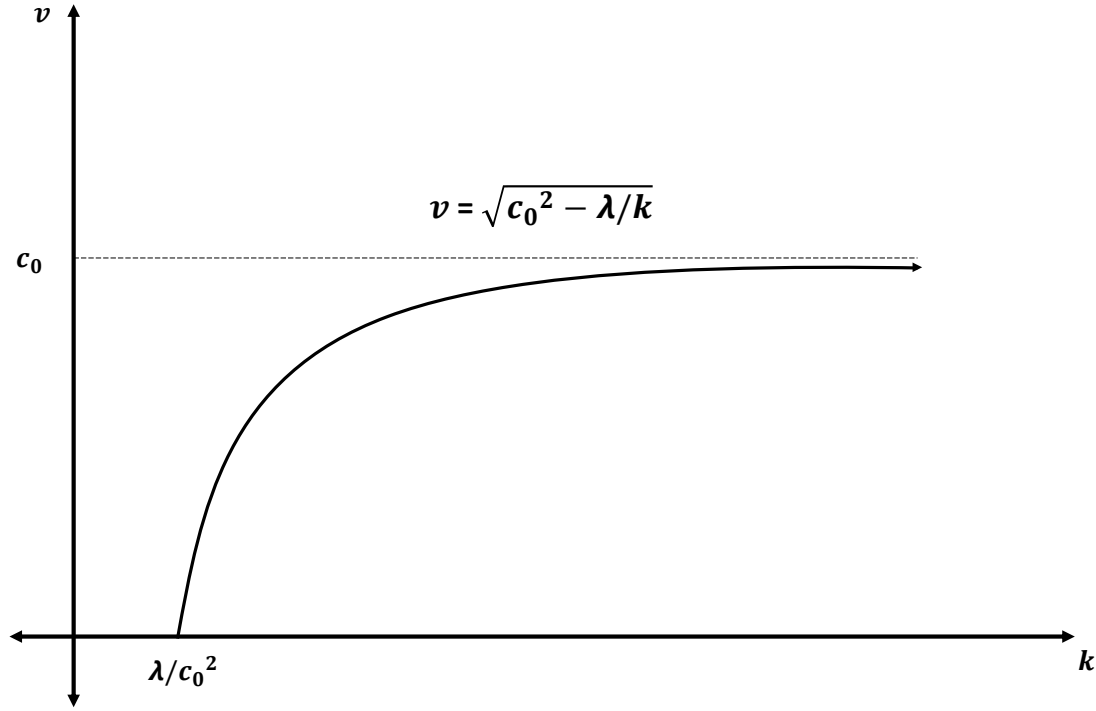


Figure 6.1: Plot of wavelength versus wave speed. For waves to propagate the condition $k > \lambda/c_0^2$ must be satisfied. Wave speed cannot exceed $c_0 = \sqrt{\frac{P_0}{\rho}}$.

6.2 Double Wave Propagation

Consider solutions of the form $u(s, t) = \psi(\xi)$, where $\xi = \omega t - ks$. This change of variables reduces equation (2.9) to

$$\omega^2 \frac{d^2 \psi}{d\xi^2} + \eta \omega \frac{d\psi}{d\xi} - 2V_0 \omega k \frac{d^2 \psi}{d\xi^2} + V_0^2 k^2 \frac{d^2 \psi}{d\xi^2} = \frac{P_0}{\rho} \left[-k \frac{d}{d\xi} \left(\frac{-k\psi'}{\sqrt{1 - k^2|\psi'|^2}} \right) \right] + i\lambda \frac{d\psi}{d\xi}. \quad (6.5)$$

the nonlinearity becomes

$$\frac{d}{d\xi} \left(\frac{\psi'}{\sqrt{1 - k^2|\psi'|^2}} \right) = \frac{d^2 \psi}{d\xi^2} + \frac{k^2}{2} \frac{d}{d\xi} [\psi' |\psi'|^2].$$

The governing equation is given by

$$\omega^2 \psi'' + \eta \omega \psi' - 2\omega k V_0 \psi'' + V_0^2 k^2 \psi'' = i\lambda k \psi' + \frac{P_0}{\rho} \psi'' + \frac{P_0 k^2}{\rho} \frac{d}{d\xi} [\psi' |\psi'|^2]. \quad (6.6)$$

Let $B_0 = \omega^2 - 2\omega k V_0 + V_0^2 k^2 - \frac{P_0}{\rho}$ and consider the linear case with $\eta = 0$. The governing equation becomes

$$B_0 \psi'' - i\lambda k \psi' = 0. \quad (6.7)$$

Traveling wave solutions of equation (6.7) exist and have the form

$$\psi(\xi) = A_2 + A_1 e^{i \frac{\lambda k}{B_0} (\omega t - k s)}, \quad (6.8)$$

for constants A_1 and A_2 .

Now consider solutions to equation (2.11) of the form $u(s, t) = u_0 e^{i(\omega t - k s)}$, with $\eta = 0$. The equation reduces to the form

$$-\omega^2 + 2V_0 \omega k - k^2 V_0^2 + \frac{P_0}{\rho} k^2 + \lambda k = 0. \quad (6.9)$$

Computing the wave speed shows the propagation of two waves with speed

$$\begin{cases} v_1 = V_0 + \sqrt{\frac{P_0}{\rho} + \frac{\lambda}{k}}, \\ v_2 = V_0 - \sqrt{\frac{P_0}{\rho} + \frac{\lambda}{k}}. \end{cases} \quad (6.10)$$

When $V_0 = 0$, there is one wave with speed $v = \sqrt{c_0^2 + \frac{\lambda}{k}}$. This corresponds to the wave speed in equation (6.4), but with traveling wave solutions given by $u(s, t) \sim u_0 e^{i(\omega t - k s)}$. Figure (6.2) displays the wave speeds as functions of the wavelength.

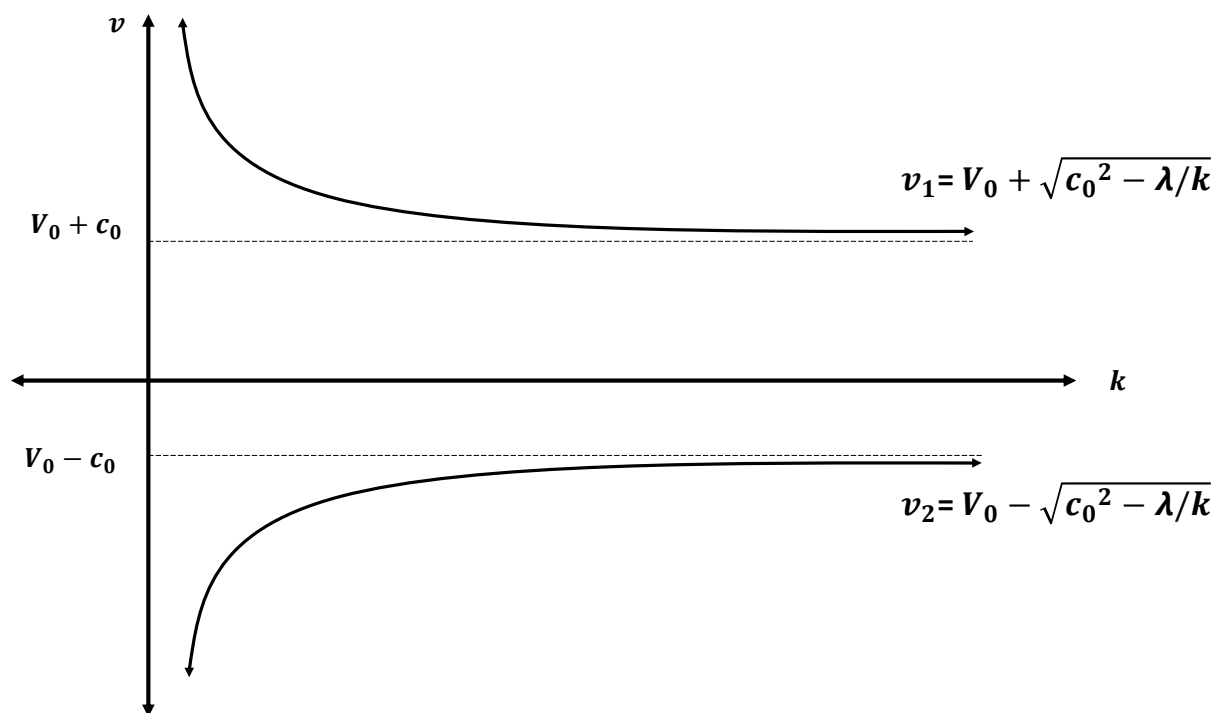


Figure 6.2: Plot of wavelength versus wave speed. Two waves are observed traveling in the same direction, but at different speeds. Notice here we consider $V_0 < c_0$.

Chapter 7

Conclusion

Stability of lateral vibrations of a moving current-carrying conductive string in a homogeneous magnetic field has been considered in this thesis. A nonlinear governing equation describing the motion of the string was derived. Static and dynamic cases were investigated. A critical value for stability of the static case has been found when uniform (along a string line) motion of a string is unstable. This critical value depends on the speed of the motion, the applied magnetic field, and the electrical current. The condition of dynamic instability is also obtained. The instability conditions for the linear and dynamic static cases do not coincide.

Applying a Galerkin Orthogonalization procedure, a nonlinear ordinary differential equation has been obtained for the amplitude of the nonlinear vibrations. The equilibrium points of this equation have been found, and the behavior of the solution in the vicinity of these points has been investigated. Nonlinear resonance vibrations of a moving string have been investigated when a non-stationary period current flows through the string. For an approximate solution the Method of Multiple Scales has been applied. Resonant solutions have been derived from which the boundaries of the unstable regions have been obtained. Domains of parameters have been defined when the string falls into pre-chaotic motion (i.e. the frequency of vibrations is doubled).

Traveling wave solutions were used to compute the wave speeds in certain cases of the general system. Certain cases considered were shown to produce no waves, indicating a dead-zone for wave propagation. Another case showed double-wave propagation is produced when the string is perturbed.

Appendices

Appendix A

The Wave Equation

As a preliminary step towards the governing equation derived in this thesis, we study the vibrations of an elastic string fixed at two points (a distance ℓ apart) with the following assumptions:

1. The string is homogeneous, i.e., the mass per unit length is constant,
2. The magnitude of the gravitational force is negligible to the tension force,
3. Each particle moves strictly vertically.

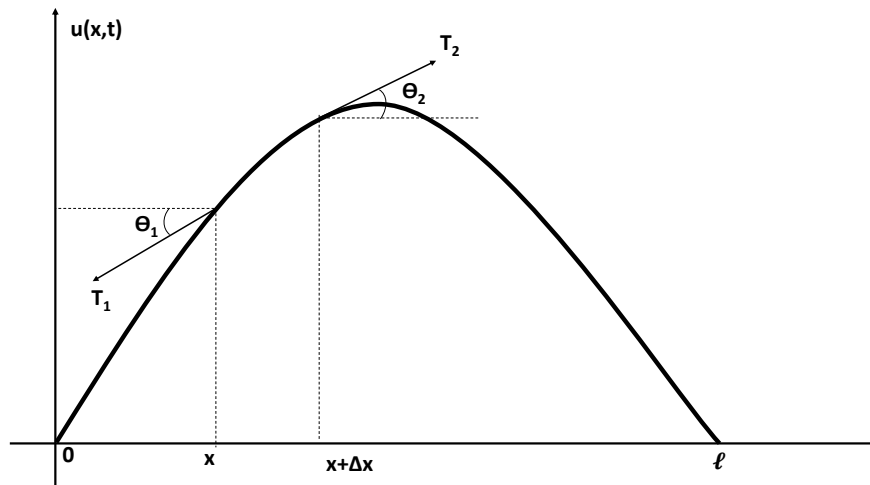


Figure A.1: Vibration of a string between two fixed points.

Let $u(s, t)$ describe the deflection at any point x at time $t > 0$. Consider forces acting on a small portion of the string, shown in figure A.1. T_1 and T_2 describe the tension forces acting at the two points $u(x, t)$ and $u(x + \Delta x, t)$, respectively. Since there is no horizontal motion, the tension in the horizontal direction is constant

$$T = T_1 \cos \theta_1 = T_2 \cos \theta_2 = \text{constant}. \quad (\text{A.1})$$

Using Newton's second law, we only have forces in the vertical direction described by

$$T_2 \sin \theta_2 - T_1 \sin \theta_1 = \rho \Delta x \frac{\partial^2 u}{\partial t^2}, \quad (\text{A.2})$$

where $\rho \Delta x$ is the mass, and $\frac{\partial^2 u}{\partial t^2}$ is the acceleration. Notice in this system,

$$\tan \theta_1 = \left(\frac{\partial u}{\partial x} \right) \Big|_x \quad \text{and} \quad \tan \theta_2 = \left(\frac{\partial u}{\partial x} \right) \Big|_{x+\Delta x}. \quad (\text{A.3})$$

Using equations (A.1)-(A.3) and taking $\Delta x \rightarrow 0$, we obtain the one-dimensional wave equation

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}, \quad (\text{A.4})$$

where $c^2 = \frac{T}{\rho}$. Equation (A.4) is considered with boundary conditions

$$u(0, t) = u(\ell, t) = 0. \quad (\text{A.5})$$

Similar to our investigation in Chapter 3, we can compute the eigenvalues. Through separation of variables, we can compute the general solution of (A.4)-(A.5) and eigenvalues, which are $\lambda_n = \frac{cn\pi}{\ell}$, $n = 1, 2, \dots$. The buckling nodes are shown in figure (A.2).

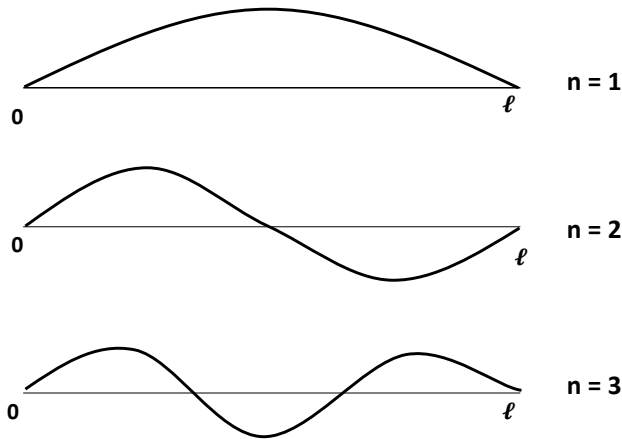


Figure A.2: Normal nodes of the string.

Appendix B

Derivation of String Equilibrium States following P. Wolfe's Approach

In [18], Wolfe presents the derivation and bifurcation analysis of an elastic conductor in a magnetic field. The following is a summary of Wolfe's approach and uses notation from [1]. These results are contrasted with our results in Section 3.1.

Consider a nonlinearly elastic wire with natural length 1. Let $s \in [0, 1]$ be the material point of the string, and $\bar{r}(s)$ be the deformed position in Euclidean 3-space (spanned by $\{\hat{i}, \hat{j}, \hat{k}\}$). Let $v(s) = |\bar{r}'(s)|$ and $\bar{e}(s) = \frac{\bar{r}'(s)}{v(s)}$; note that $\bar{e}(s)$ is the unit tangent to the wire at s . The endpoints of the string are fixed at $\bar{0}$ and $b\hat{k}$, for $b \geq 1$ (in our system we assume $b = 0$). The current is given by I , and the magnetic field is given by $\bar{B} = B\hat{k}$, for constant B . The force acting on the wire due to the current and magnetic field (Laplace Force) is given by

$$\bar{f}(s) = I\bar{r}' \times B\hat{k} = Iv\bar{e} \times B\hat{k}. \quad (\text{B.1})$$

The force exerted by the material point of $(s, 1]$ on the material of $[0, s]$ is denoted by $\bar{n}(s)$. Note we assume the wire is perfectly flexible, thus $\bar{n}(s) = N(s)\bar{e}(s)$, where $N(s)$ is the tension in the wire at s . Furthermore, we assume the wire is nonlinearly elastic so the tension at s depends on the stretch $v(s)$, i.e. $N(s) = \hat{N}(v(s), s)$. The equilibrium equation (force balance) is

$$\bar{n}'(s) + \bar{f}(s) = 0. \quad (\text{B.2})$$

Wolfe's boundary value problem is to find $\bar{r}(s)$ such that

$$(N(s)\bar{e})' + Iv\bar{e} \times B\hat{k} = \bar{0}, \quad 0 < s < 1, \quad (\text{B.3})$$

$$\bar{r}(0) = \bar{0}, \quad \bar{r}(1) = b\hat{k}, \quad b > 1. \quad (\text{B.4})$$

Note that (B.3)–(B.4) is analogous to the Sturm-Liouville problem stated in the beginning of Section 3.1. The boundary condition (B.4) at the right endpoint is varied by b , whereas in section 3.1 the conditions are fixed at zero. Wolfe first considers the case of a homogeneous wire, i.e. he assumes the $\hat{N}$ is independent of s . Then equation (B.3) becomes

$$(N(s)\bar{e})' + N(s)\bar{e} + \lambda v(\bar{e} \times \hat{k}) = \bar{0} \quad (\text{B.5})$$

where

$$\lambda = IB > 0. \quad (\text{B.6})$$

Wolfe shows that $\bar{r}(s) = bs\hat{k}$ is a trivial solution and derives states the following theorem

Theorem Suppose $N = \hat{N}(v)$ (homogeneous string). Let $\lambda > 0$, and $v > b$ satisfy

$$v\lambda/N = 2\pi m, \text{ for } m = 1, 2, \dots \quad (\text{B.7})$$

Then (B.3) and (B.4) has solutions

$$\begin{aligned} \bar{r}_{\theta_0}(s) = (N/\lambda)[\sin(\theta_0 + v\lambda/N)s - \sin \theta_0] \sin \psi \hat{i} \\ + (N/\lambda)[\cos \theta_0 - \cos(\theta_0 + v\lambda/N)s] \sin \psi \hat{j} + v \cos \psi s \hat{k}, \end{aligned} \quad (\text{B.8})$$

where $0 \leq \theta_0 \leq 2\pi$ and $\psi = \arccos(b/v)$.

Next, he considers the linear (static) case for this system. Similar to our results at the end of Section 3.1, Wolfe finds the eigenvalues

$$\lambda_m = 2\pi m \hat{N}(b)/b, \quad (\text{B.9})$$

and the parametric representation of the eigenfunctions given by

$$\begin{aligned}x_1^{(m)} &= c_1(1 - \cos 2\pi ms) + c_2 \sin(2\pi ms), \\y_1^{(m)} &= -c_1 \sin 2\pi ms + c_2(1 - \cos 2\pi ms), \\z_1^{(m)} &= 0.\end{aligned}\tag{B.10}$$

Note λ_m are bifurcation points of the equilibrium solutions. It's assumed $\frac{d}{dv}(\hat{N}(v)/v) > 0$, so Wolfe presents a bifurcation diagram describing the branches of nontrivial solutions. Essentially, for

$$0 < \lambda < \frac{2\pi\hat{N}(b)}{b},\tag{B.11}$$

the conductor will have a straight-line, trivial solutions. If

$$\frac{2\pi\hat{N}(b)}{b} < \lambda < \frac{4\pi\hat{N}(b)}{b},\tag{B.12}$$

then the string will have one spiral, or buckled node in the shape of the eigenfunction corresponding to the first eigenvalue ($m = 1$). These results are compared with our linear static case in Section 3.1. Figure 3.1 contrasts our studies and assumptions.

Appendix C

Uniqueness and Energy for Certain Cases of Our System

Consider the linear governing equation (2.11) with $V_0 = 0$ and $\eta = 0$. Physically, we are neglecting the axial motion and the viscosity on the string. Equation (2.11) becomes

$$\frac{\partial^2 u(s, t)}{\partial t^2} - c^2 \frac{\partial^2 u(s, t)}{\partial s^2} + i\gamma \frac{\partial u(s, t)}{\partial s} = 0, \quad 0 \leq s \leq \ell, \quad (\text{C.1})$$

where $c^2 = \frac{P_0}{\rho}$, $\gamma = \frac{\lambda P_0}{\rho}$, and $\lambda = \frac{I_0 B_0}{P_0}$. Note that $c = \sqrt{\frac{P_0}{\rho}}$ is the wave speed of the string. We will show uniqueness of solutions, and following [19] we will show the energy of this system is bounded. This implies that the system is nonconservative, i.e. energy is dissipated over time. Consider a change of variables $s = \ell x$, then equation (C.1) becomes

$$\frac{\partial^2 u(x, t)}{\partial t^2} - c_1^2 \frac{\partial^2 u(x, t)}{\partial x^2} + i\gamma_1 \frac{\partial u(x, t)}{\partial x} = 0, \quad 0 \leq x \leq 1, \quad (\text{C.2})$$

where $c_1^2 = \frac{c^2}{\ell^2}$, and $\gamma_1 = \frac{\gamma}{\ell}$. The initial conditions are given by

$$\begin{cases} u(x, 0) = \phi(x), \\ \frac{\partial u(x, 0)}{\partial t} = \psi(x), \end{cases} \quad (\text{C.3})$$

and boundary conditions

$$\begin{cases} \alpha_1 u(0, t) + \beta_1 u_x(0, t) = f_1(t), \\ \alpha_2 u(1, t) + \beta_2 \frac{\partial u(1, t)}{\partial x} = f_2(t). \end{cases} \quad (\text{C.4})$$

We want to find the conditions on $g(x, t)$, $\phi(x)$, $\psi(x)$, $f_1(t)$, and $f_2(t)$ such that we have uniqueness for solutions of (C.2) with initial conditions (C.3) and boundary conditions (C.4).

Now we make a transformation to a real valued system. Let $u(x, t) = v(x, t)e^{H(x, t)}$. With this form, the differential equation reduces to

$$\begin{aligned} & \frac{\partial^2 v(x, t)}{\partial t^2} - c_1^2 \frac{\partial^2 v(x, t)}{\partial x^2} + 2 \frac{\partial v(x, t)}{\partial t} \frac{\partial H(x, t)}{\partial t} + \left[i\gamma_1 - 2c_1^2 \frac{\partial H(x, t)}{\partial x} \right] \frac{\partial v(x, t)}{\partial x} \\ & + \left[\left(\frac{\partial H(x, t)}{\partial t} \right)^2 + \frac{\partial^2 H(x, t)}{\partial t^2} - c_1^2 \left(\frac{\partial H(x, t)}{\partial x} \right)^2 - c_1^2 \left(\frac{\partial^2 H(x, t)}{\partial x^2} \right) + i\gamma_1 \frac{\partial H(x, t)}{\partial x} \right] v(x, t) = 0. \end{aligned}$$

Letting $H(x, t) = \frac{i\gamma_1}{2c_1^2}x + \delta(t)$, where $\delta(t)$ is arbitrary, we get

$$\frac{\partial^2 v(x, t)}{\partial t^2} - c_1^2 \frac{\partial^2 v(x, t)}{\partial x^2} + 2 \frac{\partial v(x, t)}{\partial t} \delta'(t) + \left[\delta'(t)^2 + \delta''(t) - \frac{\gamma_1^2}{4c_1^2} \right] v(x, t) = 0. \quad (\text{C.5})$$

Consider the two cases:

(i) $\delta'(t) = 0$:

$$\Rightarrow \frac{\partial^2 v(x, t)}{\partial t^2} - c_1^2 \frac{\partial^2 v(x, t)}{\partial x^2} - \frac{\gamma_1^2}{4c_1^2} v(x, t) = 0. \quad (\text{C.6})$$

(ii) $\delta'(t) = \text{constant} = \frac{\gamma_1}{2c_1}$:

$$\Rightarrow \frac{\partial^2 v(x, t)}{\partial t^2} - c_1^2 \frac{\partial^2 v(x, t)}{\partial x^2} + \eta_1 \frac{\partial v(x, t)}{\partial t} = 0, \quad (\text{C.7})$$

where $\eta_1 = \frac{\gamma_1}{c_1} > 0$.

Note equation (C.6) is the classical wave equation with damping. To show uniqueness of $u(x, t)$ one would show uniqueness of $v(x, t)$ in cases (i) and (ii). We will only consider the second case.

C.1 Uniqueness and Bounded Energy

Consider case (ii). We want to show uniqueness for the following

$$\begin{cases} \frac{\partial^2 v(x, t)}{\partial t^2} - c_1^2 \frac{\partial^2 v(x, t)}{\partial x^2} + \eta_1 \frac{\partial v(x, t)}{\partial t} = 0, \\ v(x, 0) = \phi_0(x), \\ \frac{\partial v(x, 0)}{\partial t} = \phi_1(x). \end{cases} \quad (\text{C.8})$$

We assume $-\infty < x < \infty$. Suppose $v_1(x, t)$ and $v_2(x, t)$ satisfy (C.7), and let $w(x, t) = v_2(x, t) - v_1(x, t)$. We want to show that $w(x, t) = 0$ for uniqueness. The system we are now considering is

$$\begin{cases} \frac{\partial^2 w(x, t)}{\partial t^2} - c_1^2 \frac{\partial^2 w(x, t)}{\partial x^2} + \eta_1 \frac{\partial w(x, t)}{\partial t} = 0, \\ w(x, 0) = 0, \\ \frac{\partial w(x, 0)}{\partial t} = 0. \end{cases} \quad (\text{C.9})$$

Let $L(w) = \frac{\partial^2 w}{\partial t^2} - c_1^2 \frac{\partial^2 w}{\partial x^2} + \eta_1 \frac{\partial w}{\partial t} = 0$. Now we construct the following integral as in [19],

$$E = \int_{-\infty}^{\infty} \left[L(w) \frac{dw}{dt} \right] dx = 0. \quad (\text{C.10})$$

Note the following

$$\begin{aligned} \frac{\partial w}{\partial t} \frac{\partial^2 w}{\partial t} &= \frac{1}{2} \left(\frac{\partial w}{\partial t} \right)^2, \\ \int_{-\infty}^{\infty} \frac{\partial w}{\partial t} \frac{\partial^2 w}{\partial x^2} dx &= -\frac{1}{2} \int_{-\infty}^{\infty} \frac{\partial}{\partial t} \left(\frac{\partial w}{\partial t} \right)^2 dx, \end{aligned}$$

and

$$v_i(\pm\infty, t) = 0, \quad \frac{\partial v_i(\pm\infty, t)}{\partial t} = 0, \quad i = 1, 2.$$

The governing equation in system (C.9) becomes

$$E = \frac{1}{2} \frac{\partial}{\partial t} \int_{-\infty}^{\infty} \left[\left(\frac{\partial w}{\partial t} \right)^2 + c_1^2 \left(\frac{\partial w}{\partial x} \right)^2 \right] dx + \int_{-\infty}^{\infty} \eta_1 \left(\frac{\partial w}{\partial t} \right)^2 = 0. \quad (\text{C.11})$$

This is true only when $w(x, t) = \text{constant}$, but we need w to satisfy the initial conditions, thus $w(x, t) = 0$ and $v_1(x, t) = v_2(x, t)$. This proves uniqueness.

Next we show that the energy is bounded. Consider equation (C.7) for $0 < x < l$ with boundary conditions $v(0, t) = 0$, and $v(1, t) = 0$. Note that we can show the same result for $-\infty < x < \infty$.

Define the energy as

$$\tilde{E}(t) = \frac{1}{2} \int_0^l \left[\left(\frac{\partial v}{\partial t} \right)^2 + c_1^2 \left(\frac{\partial v}{\partial x} \right)^2 \right] dx, \quad (\text{C.12})$$

where

$$\tilde{E}(0) = \frac{1}{2} \int_0^l \left[\phi_1(x)^2 + c_1^2 \left(\frac{\partial \phi_0(x)}{\partial x} \right)^2 \right] dx < M < \infty \quad (\text{C.13})$$

for some number M .

Theorem

If $\phi_0(x)$ and $\phi_1(x)$ satisfy (C.12), then $\tilde{E}(t) \leq \tilde{E}(0)$, $(t \rightarrow \infty)$.

Proof

$$\begin{aligned} \frac{d\tilde{E}}{dt} &= \frac{1}{2} \frac{\partial}{\partial t} \int_0^l \left[\left(\frac{\partial v}{\partial t} \right)^2 + c_1^2 \left(\frac{\partial v}{\partial x} \right)^2 \right] dx = \int_0^l \left[\frac{\partial v}{\partial t} \left(\frac{\partial^2 v}{\partial t^2} \right) + c_1^2 \frac{\partial v}{\partial x} \frac{\partial^2 v}{\partial x \partial t} \right] dx \\ &= \int_0^l \left[\frac{\partial v}{\partial t} \frac{\partial^2 v}{\partial t^2} - c_1^2 \frac{\partial v}{\partial t} \frac{\partial^2 v}{\partial x^2} \right] dx = \int_0^l \frac{\partial v}{\partial t} \left(-\eta_1 \frac{\partial v}{\partial t} \right) dx = -\eta_1 \int_0^l \left(\frac{\partial v}{\partial t} \right)^2 dx. \end{aligned} \quad (\text{C.14})$$

Since $\frac{d\tilde{E}}{dt} < 0$, as $t \rightarrow \infty$, $\tilde{E}(t) < \tilde{E}(0)$.

These results show that for certain cases, the linear governing equation (2.11) has unique solutions and the energy of the system is dissipative.

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